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Uncovering plaque-glia niches in human Alzheimer's disease brains using spatial transcriptomics

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Abstract

Background Amyloid-beta (A β) plaques and their associated glial responses are hallmark features of Alzheimer's disease (AD), yet their interactions within the human brain remain poorly defined.

Methods We applied spatial transcriptomics (ST) and immunohistochemistry (IHC) to 78 postmortem brain sections from 21 individuals in the Religious Orders Study and Memory and Aging Project (ROSMAP). We paired ST with histological data and stratified spots into major categories of plaque-glia niches based on A β , GFAP, and IBA1 intensity. Leveraging published ROSMAP single-nucleus RNA-seq data, we examined differences in gene expression, cellular composition, and intercellular communication across these niches. Neuronal and glial changes were validated by IHC and quantitative analyses. We further characterized glial responses using gene set enrichment analysis (GSEA) with known mouse glial signatures and human AD-associated microglial states. Finally, we used iPSC-derived multicellular cultures and single-cell RNA sequencing (scRNA-seq) to identify cell types that, upon short-term A β exposure, recapitulate the glial responses observed in the human spatial data.

Results Low-A β regions, enriched for diffuse plaques, exhibited transcriptomic profiles consistent with greater neuronal loss than high-A β regions. High-glia regions showed increased expression of inflammatory and neurodegenerative pathways. Spatial glial responses aligned with established gene modules, including plaque-induced genes (PIGs), oligodendrocyte (OLIG) responses, disease-associated microglia (DAM), disease-associated astrocytes (DAA), and human AD-associated microglial states, indicating that diverse glial phenotypes emerge around plaques and shape the local immune environment. IHC confirmed elevated neuronal apoptosis near low-A β plaques and greater CD68 abundance and synaptic loss near glia-high plaques. In vitro, iPSC-derived microglia—but not astrocytes—exposed to A β displayed transcriptomic changes that closely mirrored the glial states identified in our ST dataset.

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Conclusions Our study provides a comprehensive spatial transcriptomic dataset from human AD brain tissue and bridges spatial gene expression with traditional neuropathology. By integrating ST, snRNA-seq, and human multicellular models, we map cellular states and molecular events within plaque-glia niches. This work offers a spatially resolved framework for dissecting plaque-glia interactions and reveals new insights into the cellular and molecular heterogeneity underlying neurodegenerative pathology.

Keywords Spatial Transcriptomics, Amyloid-beta, Glia, Alzheimer's Disease, Human Brain

Background

A defining pathological feature of Alzheimer's disease (AD) is the accumulation of amyloid-beta (A β) plaques. Microglia and astrocytes frequently cluster around these plaques, playing a crucial role in A β processing and plaque dynamics [1]. In AD animal models, glial recruitment and activation near A β plaques are commonly linked to local inflammatory responses, though their precise role, whether neuroprotective, reparative, or neurotoxic, remains debated [2, 3]. In postmortem human brains, high-resolution imaging reveals that microglia and astrocytes form reactive glial nets around A β plaques [4]. Furthermore, the glial response near plaques continues to evolve even after A β accumulation plateaus [5]. Notably, glial cells in human AD brains exhibit transcriptional signatures distinct from those in mouse models [6–8]. This raises a critical question: how do glial cells interact with A β plaques within plaque-glia niches (defined as local microenvironments containing both A β and reactive glia) to influence local cellular composition and gene expression in the human brain?

Recent omics studies have identified genes and gene modules associated with A β plaques at both bulk and single-cell/nucleus levels [6–11]. The next step in advancing the field is to link local transcriptomic differences with tissue pathology in a spatially resolved context. Advances in spatial profiling technologies now allow genome-wide molecular measurements while preserving the spatial organization of cells and pathological features. While these technologies have been applied in AD research, primarily in mouse models [12, 13], to reveal gene expression and cell composition changes near pathologies, studies on human AD brains remain limited. A recent spatial transcriptomics (ST) study in postmortem human brains identified altered co-expression patterns of gene modules near AD pathologies, but this study only comprised a handful of samples [14].

Here, we apply ST and immunohistochemistry (IHC) to 78 postmortem human brain sections from 21 individuals to obtain spatially resolved transcriptomic profiles alongside pathology measurements. With this large and unique dataset, we aim to decipher the interactions between A β plaques and glial cells by addressing two key questions: (1) What transcriptomic responses

are associated with A β plaques in human brains? (2) How do plaque-associated glia cells influence these responses? By leveraging A β , GFAP, and IBA1 intensity measurements, we stratify ST spots into major types of plaque-glia niches, uncovering distinct differences in gene expression, cellular composition, and intercellular communication. We further validate the apoptotic and neuronal effects of A β and glial activation using IHC in postmortem brain sections. Our findings reveal exacerbated synaptic and neuronal loss in low-A β or high-glia areas, indicating the diverse roles of glial cell states in shaping the local immune response in AD.

Methods

Tissue collection for spatial transcriptomics

The human brain tissues were obtained through the ROSMAP (Religious Orders Study and Memory and Aging Project) cohort studies at the Rush Alzheimer's Disease Center, Rush University Medical Center, Chicago, IL. Written informed consent was given by the donors for brain autopsy and for the use of material and clinical data for research purposes, in compliance with national ethical guidelines. Clinicopathological information of the donor, including postmortem time, age, and sex, is recorded (Supplementary Table 1). Neuropathological diagnosis of the A β deposition, neurofibrillary tangles, and other pathologies is provided (Supplementary Table 1). We selected 13 individuals clinically diagnosed as AD with high pathology and 8 controls with no cognitive impairment (NCI) and no or very little pathology. All individuals are female and matched for age. We chose all female individuals because there were insufficient males in the ROSMAP cohort matching our selection criteria (e.g. RIN > 5, age-matched, minimal freezing artifacts in frozen DLPFC). Frozen brain blocks of the right-hemisphere DLPFC from 21 individuals were cut (~ 1 cm³), transferred on dry ice, and stored at –80C.

Frozen brain blocks were embedded in OCT and cryosectioned coronally to a thickness of 10 μ m using a Leica CM1950 cryostat set at –17C. Sectioning was done in sets of three (middle section for ST and two adjacent sections for IF), with 10–50 μ m between each

set. Sections were attached to Visium slides and stored at -80°C in a slide mailer for up to one week before proceeding with spatial transcriptomics. ST-adjacent sections were attached to pre-labeled Leica superfrost plus slides and stored at -80°C in slide mailers for up to two weeks before proceeding with immunohistochemistry.

Spatial transcriptomics

Spatial Transcriptomics was performed according to the Visium Spatial Gene Expression User Guide (10X Genomics; CG000239). Briefly, cryosectioned tissue on Visium slides was transferred from -80°C to the lab on dry ice and quickly thawed for 1 min on a heat block at 37°C , then immediately transferred to methanol pre-equilibrated to -20°C and fixed at -20°C for 30 min. Sections were then stained with hematoxylin and eosin, and brightfield images were acquired with a Nikon Ti2 with NIS-Elements AR 5.11.01 64-bit software. After imaging, tissues were immediately permeabilized in 0.1% pepsin diluted in 0.1 M HCl for 18 min at 37°C . Reverse transcription, second-strand synthesis, cDNA amplification (15 cycles), and library construction were performed according to the manufacturer's instructions. For the first four individuals (#2, 8, 12, and 18), protocol version 'Rev A' was followed, and the sample index PCR was 12 cycles. For the remaining samples, protocol version 'Rev D' was followed, and the sample index PCR was 13–14 cycles. Library concentration was quantified by Qubit 1× dsDNA HS using a SpectraMax M3 plate reader. Library concentration and size distribution were assessed by Fragment Analyzer (Agilent). Sequencing was performed in three batches on an Illumina NovaSeq, targeting a minimum of 50,000 raw reads per spot.

Immunofluorescence assay of ST-adjacent sections

We first attempted to perform IHC staining on the ST section, following 10X Genomics recommendations. However, due to the modified protocol required to preserve RNA integrity (i.e., methanol instead of PFA fixation, no antigen retrieval, shorter blocking, and antibody incubations), we were unable to obtain adequate staining quality. We decided instead to perform IHC on two ST-adjacent sections. Cryosectioned tissue on Leica superfrost slides was transferred from -80°C to the lab on dry ice and quickly thawed for 1 min on a heat block at 37°C , then immediately transferred to a slide-mailer filled with PFA pre-equilibrated to 4°C and fixed at 4°C for 20 min. Slides were washed twice with DPBS before performing antigen retrieval for 10 min @ 95°C in 10 mM sodium citrate buffer, pH 6.0. Following antigen retrieval, slides were washed three times with DPBS, a hydrophobic barrier was drawn surrounding the tissue (two sections per slide), and

then sections were blocked for 1 h at RT in blocking buffer (DPBS containing 5% normal goat serum, 1% BSA, 0.1% Tween-20, and Human TruStain FcX (Fc blocker)). Sections were then incubated overnight at 4°C with primary antibodies to detect A β (4G8), GFAP, and IBA1 (diluted in blocking buffer). The next day, the primary antibody solution was removed, and sections were washed three times with DPBS (5 min per wash). Sections were then incubated for 1.5 h at RT in the dark with fluorophore-conjugated secondary antibodies (Alexa Fluor 488 for A β , Alexa Fluor 790 for GFAP, and Alexa Fluor 594 for IBA1) diluted 1:500 in blocking buffer. The secondary antibody solution was removed, and sections were washed four times with DPBS (5 min per wash), then incubated with DAPI to stain nuclei and Trueblack (according to the manufacturer's instructions) to quench tissue autofluorescence. Slides were coverslipped and imaged on the ThermoFisher CellInsight CX7 LZR High Content Analysis Platform (HCA; 100 16-bit images per section).

Immunostaining quantification for ST-adjacent sections

H&E images were acquired from ST sections with an Eclipse Ti2-E microscope (Nikon) at $20\times$ magnification in 9×9 tile tiff files with 10% overlap and were stitched into a single image per section with Adobe Photoshop (23.4.2 release). We note that some patches not covered by tissue could not be stitched, but tissue-covered areas were unaffected. To verify this, we confirmed alignment of spot coordinates to the H&E image. Fluorescent A β , GFAP, IBA1, and DAPI images were acquired from the two adjacent sections with the ThermoFisher HCA at $20\times$ magnification in 10×10 tile 16-bit tiff files. Background and shading correction were performed with the ImageJ plugin BaSiC [15]. The tiles were then stitched with a 1% overlap with the stitching plugin in ImageJ. The adjacent IHC sections were aligned to the middle/sequenced H&E section by setting 7–12 common landmarks between each adjacent image and the H&E one, then using the ImageJ plugin 'Landmark Correspondences'.

Due to the non-uniform intensity background across tiles and bright areas arising from artifacts such as tissue wrinkles and tears, we applied standard image processing techniques for further background correction and artifact removal. Using a structural element with a size well above that of amyloid plaques, microglia, and astrocytes, for each IF image, we performed grayscale opening and closing to extract large bright artifacts and local background inhomogeneity, which we regressed out from the image. With the artifact-corrected IHC images, we quantified the amount of amyloid, microglia, and astrocytes by averaging the intensity of pixels within each spot of the ST sections. We excluded spots near tissue borders overlapping with bright artifacts from our analysis.

Immunostaining of formalin-fixed paraffin-embedded (FFPE) tissue

FFPE cortical sections were pre-heated for 60 min at 60 °C in a dry oven and then rehydrated using a series of xylene and ethanol dilutions before rehydrating in DI water. Sections were then used for performing antigen retrieval for 20 min in boiling 10 mM sodium citrate buffer, pH 6.0. Following antigen retrieval, sections were washed with DPBS, incubated in 0.02N HCl for 20 min for peroxide blocking, washed with DPBS, and blocked for 1 h at RT in TNB blocking buffer (0.1 M Tris-HCl pH 7.5, 0.15 M NaCl and 0.5% Akoya blocking reagent; prepared as per manufacturer's instructions). Sections were then incubated overnight at 4 °C with primary antibodies diluted in the TNB blocking buffer to detect combinations of A β (4G8; 1:1000), GFAP (1:1500), IBA1 (1:200), cleaved Caspase-3 (1:250), NeuN (1:100), SYP (1:500) and/or CD68 (1:100). The following day, primary antibody solution was removed from sections, washed three times with TNT wash buffer (0.1 M Tris-HCl pH 7.5, 0.15 M NaCl and 0.05% Tween-20) (5 min per wash) and incubated for 1 h at RT in the dark with species-appropriate fluorophore-conjugated secondary antibodies (Alexa Fluor 568, Alexa Fluor 647, Alexa Fluor 488 or Alexa Fluor 790) diluted 1:1000 in DPBS. For cleaved Caspase-3 staining, the antibody was detected with a Mouse IgM-HRP secondary, incubated for 1 h at RT, followed by a 5 min incubation with tyramide Cy5. After secondary antibody incubation, sections were washed three times with TNT wash buffer (5 min per wash), then incubated with Trueblack for 1 min to quench tissue autofluorescence. Slides were washed two times with DPBS (5 min per wash), incubated with Hoechst 34580 (1:2000 in DPBS) for staining nuclei, mounted using Prolong antifade mountant and coverslipped. Slides were dried overnight at RT and imaged on the ThermoFisher HCA (200–300 16-bit images per section).

Immunostaining quantification for FFPE tissue sections

IHC stained slides were imaged on the ThermoFisher HCA at 20x. 24 fields per section were selected to profile the thickness of the cortex from the meninges to the white matter for each subject. Images were processed using the compartmental analysis module in the HCA software. After background correction, 4G8 staining was used to identify plaque-like objects with the following parameters: objectarea > 20 μ m, smoothing = 2. The 4G8-based objects were used as a mask to generate a region of interest (ROI) for the GFAP and IBA1 channels that encompassed both the object area and an extension from the edge of each object up to 25 μ m without overlap. Calculated metrics, including the average ROI intensity for each channel, object size, and object shape,

and counts were exported from the software. The data was filtered for an average 4G8 intensity > 2000. To adjust for technical differences in staining intensities between slides, the intensity data was scaled and centered for each case, and these resulting z-scores were used for subsequent plots. To classify plaques, an investigator blinded to the HCA measurements manually scored all 4G8-based objects as diffuse, compact, or dense core plaques (or as a staining artifact or vessel A β , both of which were excluded) yielding a final sample size of n = 722 plaques from n = 9 cases. For target intensity measurements, either the compartmental or colocalization analysis modules of the CellInsight system were used to define secondary objects and extract the average intensity and counts within each ROI. The top 25% of 4G8-based objects based on the scaled average 4G8 intensity was defined as high A β and the lower 75% as low A β . Objects with an ROI scoring in the top 25% for either scaled GFAP or scaled IBA1 average intensities were defined as glia-high and the remaining objects as glia-low. For NeuN (n = 10)/SYP (n = 9) (Fig. 5H), and CD68 (n = 10) (Fig. S7C) data, intensity and count measurements were collected from the ROI defined as the 25 μ m ring around the plaque (not including the plaque area). For cleaved Caspase-3 (n = 10) data (Fig. 3H-I), the ROI was defined as either the nuclei + or NeuN + areas within each 4G8 object and a 25 μ m ring around it. The average number of 4G8 based objects analyzed per case were 933 for SYP, 507 for NeuN, 1136 for CD68, and 517 for Caspase-3. To contrast the effects of low vs. high A β on apoptosis, we applied a negative binomial mixed model [NEBULA [16]] with Caspase-3 count as the response, NeuN + nuclei count as the offset, age at death and sex as fixed effect confounding factors and modeled within-subject correlations among ROIs from the same subject with a random effect. We also tested all nuclei count (DAPI +) as the offset. To test the effects of glia-high vs. glia-low on neuron loss, we further applied this model with NeuN count as the response. Both nuclei count and ROI area (excluding plaques) were separately tested as the offset. Finally, we examined glia effects on synapse loss and inflammation by applying linear mixed models with average SYP and CD68 intensity, respectively, as the response, age at death and sex as fixed effect confounding factors and modeled within-subject correlations among ROIs from the same subject with a random effect.

Spatial transcriptomics sequencing data processing

Sequencing data were pre-processed with Space Ranger pipeline (10 $\times$ Genomics), which mapped the barcodes against the human genome (Gencode 27), generated FASTQ files, and generated a count matrix for each section. The individual files were then merged in a Seurat object

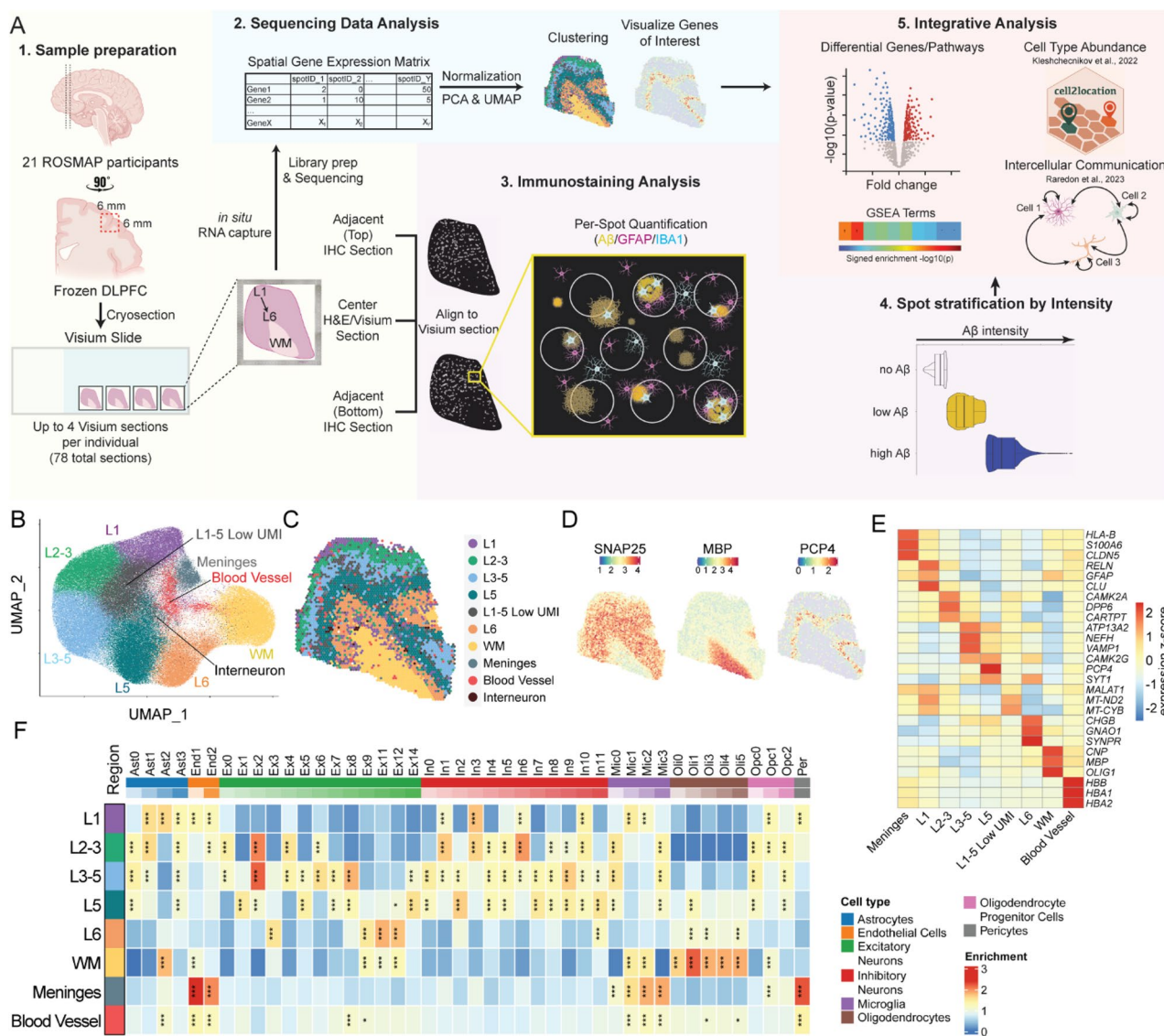


Fig. 1 Spatial Transcriptomics Profiles of Postmortem DLPFC from 21 individuals. **A** Diagrammatic summary of experimental workflow. Per individual, up to four 10-μm frozen sections of DLPFC were subjected to spatial transcriptomic sequencing using Visium. For each ST section, two adjacent sections were stained for DAPI, Aβ, GFAP, and IBA1 and imaged by fluorescence microscopy. IHC images were then aligned to the middle ST section, enabling spot-level quantification of IHC intensity for each stain and subsequent identification of genes associated with amyloid plaques and/or glial reactivity. Cartoon graphics created with BioRender.com **(B)** UMAP visualization of all ST spots from all individuals. Clusters were annotated manually based on the enriched expression of known layer- and/or cell-type-specific genes. **C** Spot clusters overlaid on a representative section (Sect. 8_C). **D** The expression level of *SNAP25* (pan-neuronal), *MBP* (oligodendrocytes; white matter), and *PCP4* (Cortical layer 5) overlaid on a representative tissue section (Sect. 8_C). **E** Heatmap showing z-score of average expression of a subset of cluster-enriched genes. **F** Heatmap of GSEA normalized enrichment score (NES) for cell2location predicted neural sub-cell-types abundance across all spots comprising each spot cluster (BH-FDR: * $p < 0.01$, ** $p < 0.001$, *** $p < 1e-4$)

(Seurat v4.2.0) and were processed in R. We calculated per-spot quality metrics and spots with low-measured genes (< 500) were removed. The Seurat::SCTransform function was used to normalize and scale the UMI counts based on regularized negative binomial regression. Principal component analysis (PCA) was performed on the top 3000 variable genes. Using the top 50 PCs (Supplementary

Fig. 1D) and accounting for technical and biological covariates such as age, RNA integrity number (RIN), library batch, and donor, we integrated the datasets using the Harmony package (v0.1.0). We explored various tools for data integration and chose Harmony for its high performance, usability, and scalability. We only used Harmony integration to identify clusters corresponding to brain regions.

All downstream analyses (differential gene expression, cell proportion estimates, and ligand-receptor analysis) were performed on pre-integration count data. After visualizing the standard deviation attributed to each Harmony embedding as an elbow plot (Supplementary Fig. 1E), we decided to use the top 10 Harmony embeddings. We applied `Seurat::runUMAP` to compute UMAP (uniform manifold approximation and projection) dimension reduction and `Seurat::findNeighbors` and `Seurat::findClusters` to identify clusters based on a shared nearest neighbor (SNN) clustering algorithm (Louvain clustering with resolution set to 0.3; all other parameters default). This resulted in 11 clusters, which were annotated based on their canonical markers and `spatialLIBD` (v7.5) to the closest anatomical cortical layer (L1, L2-3, L3-5, L5, L6), white matter (WM), meninges, and specific cellular subpopulations, namely cells with high expression of hemoglobin genes ('Blood Vessel') and *SST/NPY* ('Interneuron'). The last identified cluster, which carried a glial signature, had very few spots and so was not included in downstream analyses.

ST analysis pipeline benchmarking

Benchmarking of data integration methods was performed using the *scib* snakemake pipeline (<https://github.com/theislab/scib-pipeline>) on 16 representative sections from 4 individuals. We compared Harmony [17], scANVI [18], fastMNN [19], ComBat [20], Seurat v3 RPCA [21], scVI [22], and BBKNN [23]. Methods were applied using either Highly Variable Genes (HVG) or the full dataset (FULL), and with (+) or without (-) data scaling. We used multiple metrics for batch correction: Principal component regression (PCR batch), Graph iLISI, Graph connectivity, and K-nearest neighbor batch effect (kBET); and biological conservation: Isolated label F1, Isolated label silhouette, and Graph cLISI. Scores were computed using the default 40/60 weighted mean of these categories. Subject identifiers were used as batch labels, while Louvain-derived clusters were used as identity labels.

Downsampling to 16 sections was performed using `Seurat::Subset` on the full dataset. Seurat pipeline was run as described above, resulting in 10 clusters that were manually annotated based on their similarity to those generated from the full dataset. BayesSpace was run with standard settings after merging all 16 sections into a single object (but with sections made spatially isolated from each other). Upon inspection of the *qPlot* (Supplementary Fig. 3C), a *q* setting (e.g. number of clusters) of 8 was deemed optimal. BayesSpace clusters were also manually annotated. To assess consistency between Louvain and BayesSpace, using spots labeled as GM by Louvain as the reference, we computed the percentage of those spots that are also labeled as GM by BayesSpace, and vice versa.

Co-expression network analysis in spatial transcriptomics

We performed co-expression network analysis using all spots and sections of the ST data. We created a pseudo-bulk gene expression matrix by summing the counts by donor. The lowly expressed genes were removed, and we kept genes with at least 1 CPM (counts per million) in all samples. Different normalization methods were tested, and TMM-voom (`edgeR_3.36.0`) [24] was applied for the final matrix. To understand the major sources of variation in the gene expression matrix, we used the `variancePartition` [25] R function (v1.24.0) to run a linear mixed model and attribute the percentage of variation in expression based on technical and biological variables. To minimize the effect of unwanted confounders, `removeBatchEffect` (Limma library v3.50.1) R function was used to adjust the expression matrix by librarybatch, RIN, age, and PMI. A total of 14,534 expressed genes from 21 donors were used as input for the *SpeakEasy* (SE v1.0) [26] consensus clustering algorithm. We built the network using the results from 100 initializations of SE and Spearman correlation, which resulted in 29 modules, 23 with at least 30 genes (meaning that 99.3% of the genes were assigned to a module). The module assignment contains a list of genes by module in Supplementary Table 4.

For module functional enrichment analysis, we applied two approaches. 1) *gprofiler* (v2.0.2.1) [27] with the *g:GOST* R function: the webserver includes databases as Gene ontology, KEGG pathways from KEGG Reactome and WikiPathways; miRNA targets from miRTarBase and regulatory motif matches from TRANSFAC; tissue specificity from Human Protein Atlas; protein complexes from CORUM and human disease phenotypes from Human Phenotype Ontology. For multiple test corrections, we used the default *g:SCS* method. 2) Fisher exact test with Bonferroni correction ($\text{Adj } p < 0.05$) for specific gene lists as described in the Results section. All 23 modules were significantly enriched with at least one function.

For reproducibility, we ran the `modulePreservation` [28] function from the WGCNA R package (v1.70-3) with our ST networks and previously published bulk RNASeq network with 478 donors. The *Zsummary* statistic was accessed, and a Kruskal Wallis test was applied for significance. Modules with $Z < 2$ were considered not preserved, between 2 and 10 preserved, and $Z > 10$ highly preserved (Supplementary Fig. 4B-C).

Spot stratifications

ST spots were classified as no, low, or high A β based on the average fluorescence intensity of A β in the corresponding 55- μm -diameter area from the two adjacent immunostained sections. Spots with $\log_2(\text{average A}\beta \text{ intensity} + 1)$ less than 2.5 were not included for

downstream analysis, as these were primarily regions not covered by tissue. Spots with $\log_2(\text{average A}\beta \text{ intensity} + 1)$ values between 2.5 and 4 were classified as no A β , between 4 and 6.5 as low A β , and above 6.5 as high A β . These cutoffs were selected based on manual scoring of 781 plaques (Fig. 2D–E). A threshold of 6.5 (between low and high A β) maximized the distinction between diffuse and compact/dense-core plaque types. Glia-high spots were defined by high $\log_2(\text{average intensity} + 1)$ of IBA1 and/or GFAP (> 6.8 for IBA1 and/or > 7.2 for GFAP), while glia-low spots had low intensities for both IBA1 and GFAP (< 6.2 for IBA1 and < 6.5 for GFAP). Manual assessment of glia-high/low spots indicated that glia-high spots typically contain > 4 glial cells (astrocytes expressing GFAP and/or microglia expressing IBA1), while glia-low spots contain ~ 0 –4 (Supplementary Fig. 7). Different glia intensity thresholds were tested, but increasing the stringency of thresholds or isolating single-positive spots (e.g. GFAP +/IBA1- or GFAP-/IBA1 +) reduced the number of sections available to generate pseudobulk expression profiles, and thus reduced the power for DEG detection.

Statistical analysis to identify A β and glia differential genes

For each brain section of a subject, we summed the UMI counts of all spots belonging to a plaque-glia niche type to generate a pseudobulk estimate for each gene. We only kept expressed genes with counts per million (CPM) > 1 in at least 80% of the brain sections. With the pseudobulk estimates, we estimated the library sizes of the brain sections using the trimmed mean of M-values (TMM) of edgeR [24] (edgeR_3.36.0) and transformed the pseudobulk estimates into $\log_2$ CPM using voom-limma (limma_3.50.1). For each plaque-glia niche type, we excluded brain sections with < 50 spots. To contrast the normalized pseudobulk expression of a given pair of plaque-glia niche types, we applied linear mixed models (LMM) with plaque-glia niche types coded as a binary variable. We accounted for age, RIN, batch, and AD status as fixed effects and modeled within-subject correlation with a random effect. We declared significance at an α of 0.05 with false discovery rate (FDR) correction across all genes and all A β and glia contrasts.

For interpretation, we applied GSEA [29] with t-values of the niche type contrasts as scores and GO terms as genesets. We also used markers of PIG [30], OLIG [30], DAM [31], and DAA [32] from mouse studies as well as different microglia states identified in human post-mortem AD brain [8] as genesets. Moreover, with our co-cultured scRNA data, we extracted marker genes or A β induced genes for iMGL, astrocytes, neurons and iMGL1-4 subtypes, which were further used as genesets.

Cell abundance estimates

The abundance of specific brain cell types was estimated for each ST spot using cell2location [16]. The average expression of 8 major cell types was estimated from DLPFC snRNA-seq data [3] and used as a reference for cell deconvolution with cell2location. We estimated the enrichment of each cell type in each UMAP cluster using GSEA. Spots in each cluster were ranked using the cell2location predicted abundance, and overrepresentation normalized enrichment scores were calculated for each cell type. *P*-values were estimated using 100 permutations and adjusted for multiple comparisons using the Benjamini–Hochberg false discovery rate (BH-FDR) correction.

Ligand-receptor analysis

Local microenvironment interactions were estimated from ST data using NICHES (Niche Interactions and Communication Heterogeneity in Extracellular Signaling v1.0.0) [33]. We used the Omnipath database (v3.2.8) as the ligand-receptor (LR) reference and used default parameters for data imputation, k-nearest neighbor graph ($k = 4$), and Niche Matrix Construction (summing signaling input from Euclidean neighbors). Each section was analyzed separately, and results were merged for downstream analysis. For stratified spot differential expression comparisons, spot-level LR matrices were first averaged by section and stratification. Only data from gray matter spots and sections with at least fifty spots per group were included. LR genes with CPM ≤ 1 in $< 50\%$ of sections were removed. Then, the aggregated mean values were normalized and transformed by quantile-voom using the *limma* R package (v3.50.3). A mixed linear model was applied to compare the normalized values between each stratification group while accounting for sections derived from the same donors using the *duplicateCorrelation* function (limma_3.50.3). *P*-values were adjusted for multiple comparisons using BH-FDR correction.

iPSC cultures and differentiation

Two iPSC lines (AICS-0090–391 purchased from the Coriell Institute and BR28 [34] obtained from Dr. Tracy Young-Pearse) were used for the culture experiments. iPSCs were maintained on Matrigel-coated dishes in mTeSR Plus (StemCell Technologies). iPSCs were passaged every 4 days with ReLeSR (StemCell Technologies) according to the manufacturer's protocol.

To generate iMGL, we followed previously published protocols [35, 36] with minor modifications using AICS-0090–391 iPSCs. iPSC colonies were seeded at a density of 20–40 per well of a Matrigel-coated 12-well plate in mTeSR Plus supplemented with 10 μM Y-27632 and

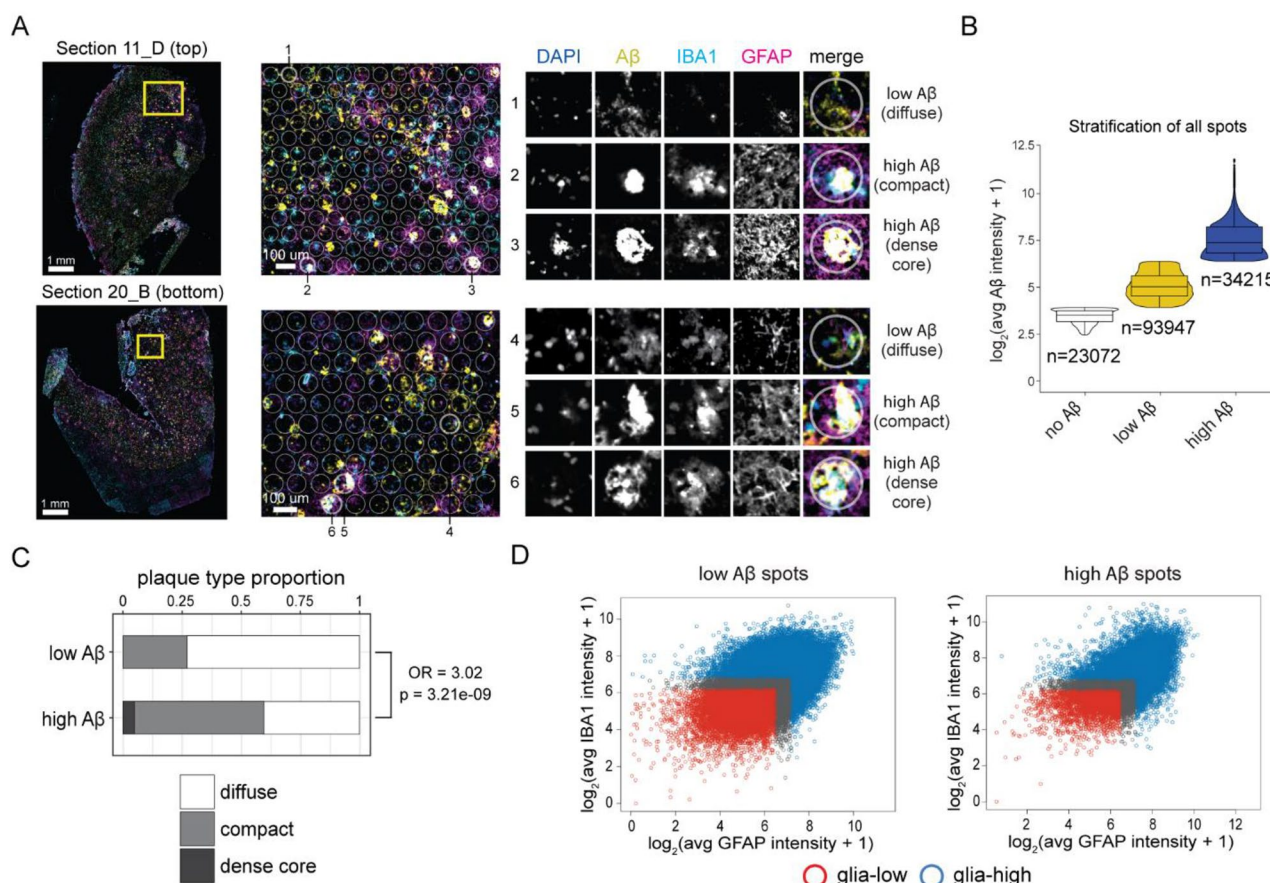


Fig. 2 Stratification of tissue microdomains by adjacent-section IHC intensities. **A** Representative images of different plaque types from two ST-adjacent sections. Six spots (1–6; including all major plaque types) are shown with a 55-μm spot area superimposed to scale. **B** We stratified all ST spots by Aβ intensity into no (white; $\log_2(\text{avg A}\beta \text{ intensity} + 1)$ between 2.5 and 4), low (yellow; $\log_2(\text{avg A}\beta \text{ intensity} + 1)$ between 4 and 6.5), or high (blue; $\log_2(\text{avg A}\beta \text{ intensity} + 1) > 6.5$) groups. The distribution of Aβ IF intensity is shown for each group. **C** A subset of 781 spots from ST-adjacent sections were selected for manual plaque-type annotation (Supplementary Table 5). The proportion of plaque types is shown, stratified by low or high intensity of Aβ [odds ratio (OR) and p-value based on Fisher's Exact test indicated]. **D** Spots were further stratified by the abundance of glia (GFAP: x-axis, IBA1: y-axis). Scatterplots show the average GFAP and IBA1 intensities for each ST spot among Low Aβ (left) and High Aβ (right) groups (red: glia-low, blue: glia-high). Gray-colored spots had intermediate GFAP/IBA1 intensity and were not sorted into a group

cultured for 48 h. At d0, colonies were fed with hematopoietic stem cell (iHPC) basal media [IMDM (50%), F12 (50%), insulin (0.02 mg/ml), ITS-G (2% v/v), L-ascorbic acid 2-Phosphate magnesium (64 μg/ml), monothioglycerol (400 μM), PVA (10 ng/ml), Glutamax (1X), chemically defined lipid concentrate (1X), and non-essential amino acids (NEAA; 1X)] supplemented with FGF2 (50 ng/ml), BMP4 (50 ng/ml), Activin-A (12.5 ng/ml), and LiCl (2 mM). At d2, cultures were fed with iHPC basal media supplemented with FGF2 (50 ng/ml) and VEGF (50 ng/ml). At d4, cultures were fed with iHPC basal media supplemented with FGF2 (50 ng/ml), VEGF (50 ng/ml), TPO (50 ng/ml), SCF (10 ng/ml), IL-6 (50 ng/ml), and IL-3 (10 ng/ml) and was replaced every other day until d12. At d12, iHPCs were gently resuspended

in MGL differentiation media [DMEM/F12, HEPES, insulin (0.2 mg/ml), ITS-G (2% v/v), B27 (2% v/v), N2 (0.5%, v/v), monothioglycerol (400 μM), Glutamax (1X), NEAA (1X), and additional insulin (5 μg/ml)] supplemented with M-CSF (25 ng/ml), IL-34 (100 ng/ml), and TGFβ-1 (50 ng/ml) and seeded at 2×10^5 cells per well of a Matrigel-coated 6-well plate. Additional MGL differentiation media was added to cultures every two days. For the remainder of the differentiation, cells were passaged as needed to maintain a density of $2\text{--}5 \times 10^4$ cells per cm^2 , retaining 50% of the media when replating. From d30–d38, MGL differentiation media was supplemented with M-CSF, IL-34, TGFβ-1, CD200 (100 ng/mL), and CX3CL1 (100 ng/mL).

To generate astrocyte/neuron cultures, we first differentiated BR28 iPSCs into neural progenitors (NPCs). iPSCs were dissociated with accutase and replated to Matrigel-coated plates at 2.5×10^5 cells/cm². From the following day, cultures were fed daily with neural induction media [DMEM/F12, 1X N2, 1X B27, 2 mM Glutamax, 100 μ M NEAA, 110 μ M BME, 0.05% BSA] supplemented with 5 μ M SB432542 and 50 μ M LDN193189 until day 12. We then used Stem Cell Technologies astrocyte differentiation and maturation media and protocol to generate mixed neuron/astrocyte cultures. Briefly, day 12 NPCs were dissociated and replated to Matrigel-coated plates in STEMdiff astrocyte differentiation media (100–0013). Cultures were fed daily until day 18, then passaged and replated at 2.5×10^5 cells/cm². From day 18 to 35, cultures were fed every other day and passaged every 6 days at 2.5×10^5 cells/cm². From day 36 through day 68, cultures were passaged and replated in STEMdiff astrocyte maturation media (100–0016). Cultures were fed every other day and passaged every 6 days at 2.5×10^5 cells/cm².

A β preparation

Amyloid-beta (A β _{1–42}) was reconstituted and aggregated according to the Fujifilm/Cellular Dynamics Labeling Amyloid Beta protocol using the amyloid-beta aggregation kit (rPeptide). Briefly, lyophilized A β was reconstituted in 250 μ L 10 mM NaOH and transferred to a 1.5 mL microcentrifuge tube. 250 μ L of HPLC water was added, followed by 56 μ L of 10X TBS (pH 7.4) and mixed by gentle pipetting. The A β solution was allowed to aggregate overnight in a normoxic incubator. The next day, the A β solution was centrifuged at 16,000xg for 1 min to collect the aggregates. The supernatant was removed, and aggregates were washed with 200 μ L 0.1 M NaHCO₃ and resuspended by pipetting. Aggregates were washed an additional 4 times in Hank's Balanced Salt Solution (HBSS) and resuspended in 200 μ L HBSS, sonicated for 10 min, and stored in individual aliquots at -20°C .

Co-culture experiments

Day 38 iMGL were collected and replated onto day 68 astrocyte/neuron cultures in STEMdiff astrocyte maturation media supplemented with M-CSF (25 ng/ml), IL-34 (100 ng/ml), TGF β –1 (50 ng/ml), CD200 (100 ng/mL) and CX3CL1 (100 ng/mL) at a ratio of 1:4. Cells were co-cultured for 72 h prior to treatment with 50 μ g/ml oligomeric-A β or vehicle. After 24 h, both adherent and suspension cell fractions were collected, washed with PBS, and dissociated with accutase. Cells were washed in wash buffer [PBS, 2% BSA, 0.2U/ μ L RNase inhibitor] and centrifuged at 300xg for 5 min at 4°C , then filtered

through 70 Flowmi strainers. Cells were counted using trypan blue on a Countess II (Invitrogen).

Single-cell RNA sequencing and data processing

We prepared separate libraries from both the adherent and suspension cell fractions using 10X Genomics 3' single-cell gene expression assays v3 and sequenced them on a NovaSeq 6000 to a depth of 50 k reads/cell. Following sequencing and FASTQ generation, raw count matrices were produced using Cell Ranger v6.0.1. The resulting count matrices were corrected for ambient RNA using CellBender [37] v0.3.0, and then processed using the Seurat v4.2.0 package in R. Cells containing <2000 or >9000 genes and/or >7.5% mitochondrial RNA reads were removed. Data was scaled and log normalized, and principal component analysis (PCA) was performed on the top 2000 variable genes. We then used Seurat to perform UMAP dimension reduction on the top 15 PCs and used Seurat::findNeighbors and Seurat::findClusters to identify clusters based on a shared nearest neighbor (SNN) clustering algorithm with neighbors = 20 and Louvain resolution = 0.3. The data was manually inspected for cell doublets, and those containing mixed cluster signatures were removed, leaving 13,740 cells. The data processing steps and clustering procedure were repeated on the cleaned dataset. To identify marker genes for each cluster, we performed a Wilcoxon Rank Sum test with Bonferroni correction using Seurat::FindMarkers, testing each population against all other populations. Clusters were then annotated using canonical markers. To find genes differentially expressed between A β -treated and control cells, we used Seurat::FindMarkers to perform a Wilcoxon Rank Sum test with Bonferroni correction to compare A β -treated cells against control cells within the clusters of interest. To further investigate iMGL, we subsetted the A β -treated iMGL and repeated the data scaling and normalization procedure using only those 3875 cells. We used Seurat to perform UMAP dimension reduction on the top 15 PCs and used Seurat::findNeighbors and Seurat::findClusters to identify clusters based on a shared nearest neighbor (SNN) clustering algorithm with neighbors = 20 and Louvain resolution = 0.3. To identify marker genes for each cluster, we performed a Wilcoxon Rank Sum test with Bonferroni correction using Seurat::FindMarkers, testing each iMGL cluster against all other iMGL populations. These cluster marker genes were used as genesets for GSEA.

Results

Spatial transcriptomics on postmortem DLPFC of 21

ROSMAP participants

We obtained 1 cm³ blocks of fresh-frozen tissue from the dorsolateral prefrontal cortex (DLPFC) of 21 female

ROSMAP participants, 13 with AD and 8 controls with no cognitive impairment (NCI) (Fig. 1A; Supplementary Table 1). We sectioned tissue coronally to acquire up to four 10- μ m sections per brain for 10X Visium ST, and for each ST section, two adjacent sections for IHC staining. For ST sections, we performed H&E staining, brightfield imaging, ST library construction, and sequencing based on the 10X protocol (Fig. 1A). In total, we obtained spatial gene expression from 258,987 spots and detected an average of 3,505 genes and 9,908 transcripts/unique molecular identifiers (UMIs) per spot (Supplementary Fig. 1A-C).

To determine the layer identities of ST spots, we first normalized the ST data using SCTransform [25] to account for technical artifacts and applied Harmony [17] to correct for sample-specific differences. For computational rigor, we compared Harmony against alternative data integration methods (including scANVI [18], fast-MNN [19], ComBat [20], Seurat v3 RPCA [21], scVI [22], and BBKNN [23]) on a subset of 16 sections. Our analysis demonstrated that Harmony is one of the top performers in terms of batch effect correction and biological conservation (Supplementary Fig. 2). Note that we used SCTransformed Harmony-integrated expression values only for spot clustering. We then performed dimensionality reduction and unsupervised clustering with the Louvain algorithm to identify clusters of spots with similar transcriptional profiles. We manually annotated the clusters based on known layer-specific marker genes (Fig. 1B, 1E; Supplementary Table 2). As expected, the spots clustered by cortical layer or tissue region (Fig. 1C), with marker genes, such as *SNAP25*, *MBP*, and *PCP4*, highly expressed in gray matter (GM), white matter (WM), and layer 5 (L5), respectively (Fig. 1D). The proportions of spots assigned to the clusters are independent of AD status and are relatively consistent across individuals (Supplementary Fig. 1F-J). The cluster-enriched genes overlap well with those found in bulk RNA-seq and other ST studies on human DLPFC [38–40] (Supplementary Fig. 1K).

Using a published single-nuclei RNA-seq dataset of the DLPFC [9] as a reference of 44 brain cell types, we further applied cell2location [41] to estimate cell type abundance in each spot (Supplementary Table 3). We found that our annotated clusters comprise the expected major cell types (Fig. 1F). For example, WM spots are enriched for oligodendrocytes; meninges and blood vessel spots are enriched for endothelial cells and pericytes; and cortical layers are enriched for neuronal subtypes (Fig. 1F). To further assess the cluster assignment of the ST spots, we repeated our pipeline, which includes SCTransform (SCT) normalization [25], Harmony integration [17], and Louvain clustering [42], on a subset of 16 sections

and compared the results against BayesSpace, a method designed for spatial clustering [43]. While BayesSpace clusters were inherently more spatially contiguous than Louvain clusters, BayesSpace did not identify clusters of spots with sporadic spatial distributions, such as Blood Vessel and Interneuron clusters (Supplementary Fig. 3C-D). Notably, 99% of spots labeled as GM by Louvain were also labeled as GM by BayesSpace (Supplementary Fig. 3G).

As a data quality check, we compared the co-expression patterns of our ST data against that of bulk RNASeq data from the DLPFC of 478 individuals [11]. Specifically, we generated pseudobulk estimates of gene expression by summing the UMIs across all spots of each individual and applied the Speakeasy algorithm [26] to cluster genes into co-expression modules (Supplementary Fig. 4A). We identified 23 modules ranging from 30 to 1,199 genes per module. All ST modules are preserved in the bulk dataset and vice versa (Supplementary Fig. 4B-C). Gene set enrichment analysis (GSEA) [29] shows that all 23 modules are enriched for relevant biological function and/or neural cell types (Supplementary Fig. 4D; Supplementary Table 4).

ST spot stratification by A β and glial staining

We collected two adjacent sections for each ST section and stained them for A β , GFAP, and IBA1. We used GFAP and IBA1 because studies show that increased expression of GFAP [44] and IBA1 [6] well correlate with astrocyte reactivity and microglia activation, respectively, near A β -amyloid plaques. After imaging the stained sections, we performed background correction using BaSIC [15] to account for field-specific technical artifacts in establishing a uniform baseline of signal intensity. We then aligned each pair of adjacent IHC images to the middle ST section and quantified the average fluorescence intensity per spot for each antibody staining. To assess the accuracy of the alignment, we correlated GFAP gene expression from ST data and protein expression from IHC data across all spots for each section (Supplementary Fig. 5A-B).

To investigate the effects of A β plaques, we first stratified the 258,986 ST spots into three groups: no, low, and high A β , based on the average fluorescence intensity of A β in the corresponding 55- μ m-diameter area from the two adjacent IHC stained sections (Fig. 2A-B, Methods). Specifically, spots with $\log_2(\text{average A}\beta \text{ intensity} + 1)$ values between 2.5 and 4 were classified as no A β (e.g. background fluorescence intensity), between 4 and 6.5 as low A β , and above 6.5 as high A β . Values below 2.5 were only observed in spots not covered by tissue. These cutoffs were selected based on manual scoring of 781 plaques. A threshold of 6.5 (between low and high A β) maximized

the distinction between diffuse and compact/dense-core plaque types. All individuals (including 8 controls) contained spots in each of the three groups, but reassuringly, controls were enriched for no/low A β while AD individuals were enriched for high A β (Supplementary Fig. 5C). Under these cutoffs, the high A β spots are more enriched for compact and dense core plaques and the low A β spots are more enriched for diffuse plaques ($OR = 3.02$, $p = 3.21e-9$, Fig. 2C). Consistently, a recent vibrational imaging study also demonstrated increasing A β contents alongside the ascending plaque stages from diffuse to compact to dense core [45]. To validate the plaque type enrichment result (Fig. 2C), we performed IHC on fixed DLPFC sections from 9 AD ROSMAP participants and manually annotated 722 plaques (Supplementary Fig. 6A). Compared to fresh-frozen sections used for ST, these FFPE sections have better preservation of tissue morphology, cleaner antibody signal, and increased image resolution. We stratified the 722 plaques based on A β intensity into low (bottom 75%) or high (top 25%) groups (Supplementary Fig. 6B). Consistent with our observation on frozen sections, we found that low-A β areas are enriched for diffuse plaques, while high-A β areas are enriched for compact and dense core plaques ($OR = 14.98$, $p = 4.89e-38$, Supplementary Fig. 6C). We then further stratified the low and high A β into glia-high spots [defined by $\log_2(\text{average intensity} + 1)$ of IBA1 and/or GFAP > 6.8 for IBA1 and/or > 7.2 for GFAP] and glia-low spots [defined by intensities for both IBA1 and GFAP < 6.2 for IBA1 and < 6.5 for GFAP]. Manual assessment of glia-high/low spots indicated that glia-high spots typically contain greater than 4 glial cells (astrocytes expressing GFAP and/or microglia expressing IBA1), while glia-low spots contain ~ 0 –4 glial cells (Fig. 2D; Supplementary Fig. 7).

Low-A β spots show transcriptomic profiles indicative of a more neurotoxic local environment

To examine the transcriptomic effects of A β , we first summed the expression values of all gray matter spots within a brain section for each stratified spot group (Fig. 3A) to generate a pseudobulk matrix for that given section. We then normalized the pseudobulk estimates across brain sections using TMM-voom [46]. To contrast the normalized pseudobulk expression of the stratified spot groups, we applied linear mixed models (LMM) with false discovery rate (FDR) correction for all tested genes and contrasts. We accounted for age, RNA Integrity Number (RIN), and batch as fixed effects and within-subject correlation as a random effect. Contrasting expression levels of low A β against high A β spots, we detected 93, 140, and 159 differentially expressed genes (DEGs) for combined, glia-low, and glia-high conditions

(FDR-adjusted $p < 0.05$; Fig. 3B–D; Supplementary Table 6). Many DEGs are unique for a specific contrast, with only 18 genes overlapping among all three contrasts. These results indicate that low A β spots display different local transcriptomes compared to high A β spots, and the amount of surrounding reactive glia modify local responses.

GSEA on the low versus high A β contrasts shows reduced synaptic functions (Fig. 3E), especially under the glia-high condition. Notably, the downregulation of intermediate filament assembly and upregulation of the ECM/cell adhesion pathways are more pronounced under the glia-low condition. By contrast, the downregulation of ion transport and upregulation of immune and apoptosis pathways are more pronounced under the glia-high condition. In addition, we observed a negative enrichment of genes associated with neurofibrillary tangles.

Furthermore, we performed GSEA with specific gene sets relevant to plaque-glia niches (Fig. 3F), including previously reported multicellular gene module of plaque-induced genes (PIG) [30], gene module changes related to oligodendrocyte (OLIG) response [30], disease-associated microglia (DAM) [31], and disease-associated astrocytes (DAA) [32] detected in mouse AD models, as well as different microglia states identified in human AD brains [8]. We detected a positive enrichment of PIG, OLIG, DAM and DAA under both glia-low and glia-high conditions, indicating some degree of preservation in response to disease pathology between humans and mice. Our result also showed a positive enrichment for OLIG signature under all glia conditions, concordant with the previous mouse ST study showing that OLIG module is highly expressed in the mild A β domains but decreases in microenvironment with dense A β pathology [30]. Among the reported human AD microglia states [8], we detected the strongest enrichment for the MG3 signature under all glia conditions, especially under the glia-high condition. MG3 microglia were reported as “ribosome biogenesis” MG state, which most resembles mouse DAM. MG3 DAM microglia also express inflammatory genes enriched for cytokine production, antigen presentation, and microglial activation [8]. Of note, we also detected an enrichment of reported MG4 (lipid processing), MG5 (phagocytic), MG6 (stress signature), MG7 (glycolytic), and MG8 (inflammatory III) for the low A β glia-high condition, suggesting that microglia activation states contribute to the differential responses between low and high A β spots.

At the gene level, we detected lower expression of neuronal (*NEFH*, *NEFM*) and synaptic (*SYT13*, *SYNGR1*, *SLC17A6*, *GABRD*) genes, and higher expression of apoptosis (*CIDEA*) genes in the low A β spots across all the glia conditions (Fig. 3G). The expression of extracellular matrix (ECM)- and immune-related genes (*ICAM1*,

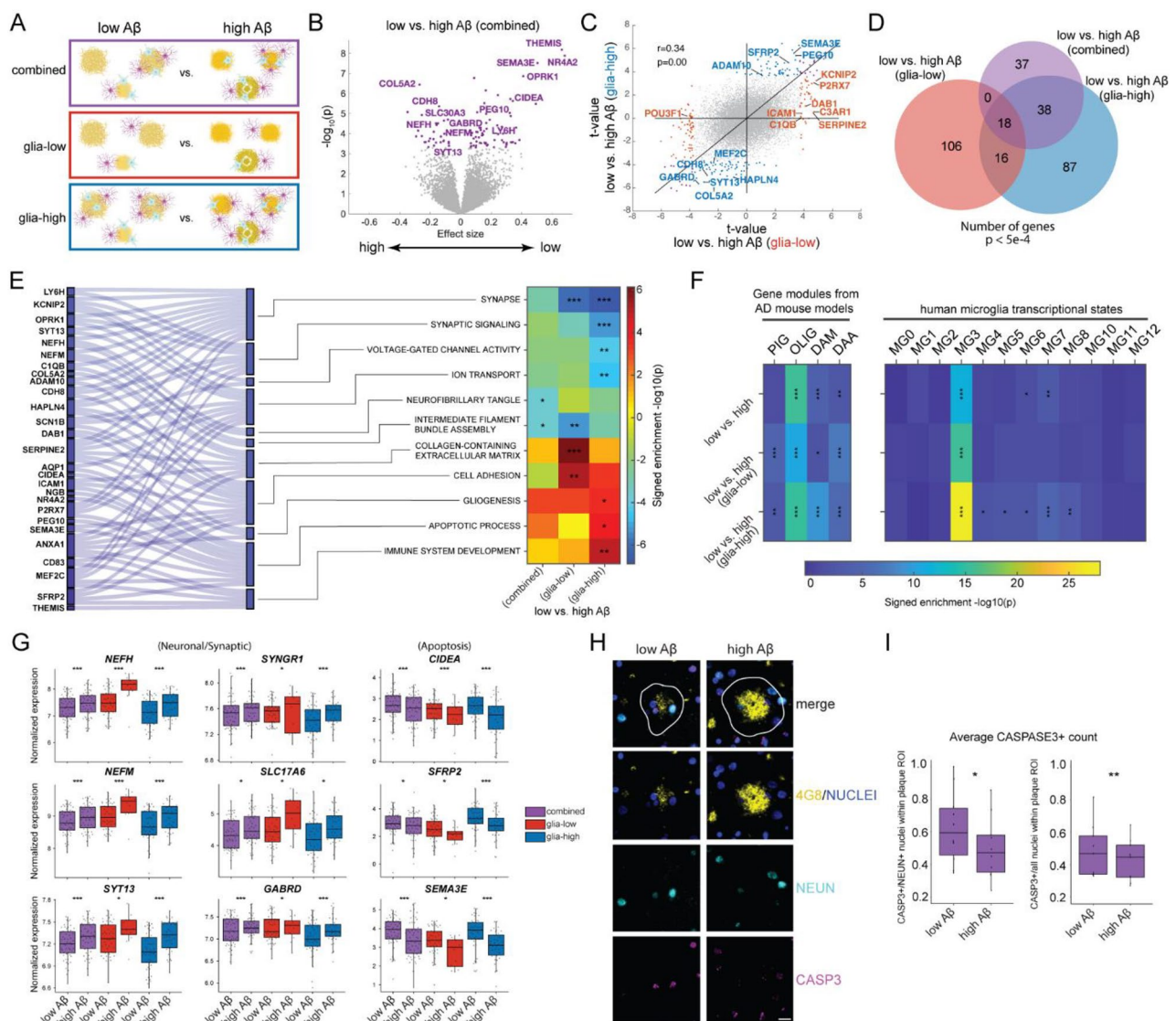


Fig. 3 Differential effects of low vs. high Aβ on the local transcriptome. **A** Schematic of the three contrasts tested. Aβ and glia graphics created with BioRender.com **(B)** Volcano plot of low vs. high Aβ. Genes with FDR-adjusted $p < 0.05$ in purple. **C** Scatterplot of Aβ effects stratified by glia-low or glia-high. Genes with FDR-adjusted $p < 0.05$ in red and blue for glia-low and glia-high conditions, respectively. **D** Venn diagram showing the number of DEGs found across Aβ contrasts. **E** GSEA enrichment of Aβ contrasts showing GO terms and canonical pathways ($*p < 5e-4$, $**p < 5e-5$, $***p < 5e-6$) **(F)** GSEA enrichment of Aβ contrasts for relevant genesets ($*p < 5e-2$, $**p < 5e-3$, $***p < 5e-4$). PIG³¹: plaque-induced genes, OLIG³¹: oligodendrocyte gene module, DAM³²: disease-associated microglia, DAA³³: disease-associated astrocytes; MG0-12: microglia states in human AD brains⁸. **G** Boxplots of representative genes for each Aβ contrast (LMM, FDR-corrected; $*p < 0.05$, $***p < 5e-6$). **H** Images of representative plaques from FFPE tissue sections stained with DAPI (blue), Aβ (4G8; yellow), NEUN (cyan), and cleaved Caspase 3 (magenta) (scale bar = 25 μm). **I** Average number of Caspase 3 puncta within NEUN + nuclei (left) or total nuclei (right) in the area within and surrounding plaques. Points represent average values from > 100 ROIs for each of 10 AD individuals (paired t-test; $*p = 0.041$, $**p = 2.61e-4$)

SERPINE2, *DAB1*) are higher in the low Aβ spots under the glia-low condition (Supplementary Fig. 8A). By contrast, the expression of apoptosis/immune-related genes (*SEMA3E*, *ADAM10*, *SFRP2*) are higher in the low Aβ spots under the glia-high condition (Fig. 3G, Supplementary Fig. 8). These results indicate that the low Aβ spots represent a more neurotoxic local environment than

high Aβ spots, and plaque-associated reactive glia modulate pathogenic mechanisms. To validate the observed differences in the apoptosis process between low and high Aβ spots, we conducted IHC on FFPE brain sections from 10 additional AD individuals with antibodies against NeuN and cleaved caspase-3. For Aβ staining, we generate a region of interest (ROI) for each 4G8

+staining object. Consistent with our ST IHC stratification criteria, the top 25% of 4G8-based objects based on the scaled average 4G8 intensity was defined as high A β and the lower 75% as low A β . We collected ~517 ROIs for cleaved Caspase-3 from those 10 AD individuals. The ROI was defined as the nuclei +/-NeuN + areas within each 4G8 object and a 25 μ m ring around it. To contrast the effects of low vs. high A β on apoptosis, we applied a negative binomial mixed model (NEBULA) with Caspase-3 count as the response, NeuN + nuclei count as the offset, age at death, and sex as fixed effect confounding factors and modeled within-subject correlations among ROIs from the same subject with a random effect. We also tested all nuclei count (DAPI +) as the offset. We detected more cleaved caspase-3 puncta co-localized with NeuN + nuclei and total nuclei near low A β plaques compared to high A β plaques (Fig. 3H-I).

Inferred intercellular communication suggests neurodegeneration in low A β spots

To examine differential A β effects on intercellular communications, we used NICHES [42], an algorithm that allows the embedding of ligand–receptor (LR) signal proxies to estimate the local microenvironment for each spot in ST data. Notably, NICHES algorithm does not restrict analysis to biochemically validated ligand–receptor interactions; rather, it identifies spatially co-expressed gene pairs between neighboring cell types as proxies of potential intercellular signaling. We first conducted low versus no A β and low versus high A β differential analysis to identify differentially expressed LRs (Supplementary Table 7). We noticed a positive association ($R = 0.85$) between these contrasts. Many LR pairs were dysregulated in both contrasts, with functions related to synapses, neuron differentiation, and ECM pathways (Fig. 4A, 4B). These results indicate that the low A β spots constitute a more neurotoxic microenvironment than high A β spots. Consistently, low A β spots show higher expression of genes related to apoptosis compared with no A β spots. In addition, low A β spots also show LRs (*APP-DCC*, *APP-VLDLR*, *APOE-LDLR*) related to greater vesicle-mediated transport compared with no A β and high A β spots (Fig. 4C), suggesting a potential compensatory mechanism for A β clearance. Our analysis suggests that low A β spots present more profound LR differences related to neuronal and synaptic degeneration than high A β spots, highlighting the importance of early intervention during plaque accumulation. In addition, low A β spots were also enriched for LRs related to greater Mitogen-Activated Protein Kinase (MAPK) cascades and immune pathways (*CIQA-CD93*, *CD99-PILRA*) compared with high A β spots. Immune modulation has

been an active topic for AD therapies. On the other hand, A β has been shown to activate MAPK, and inhibiting MAPKs reduces A β generation and alleviates cognitive impairments in AD models [43]. Our results corroborate those studies, suggesting that combining immune modulation and MAPK inhibition may provide an effective therapeutic approach to alleviate A β neuronal damage, especially in early AD.

Plaques with surrounding glia display transcriptomic differences indicative of elevated neuronal toxicity

To test whether plaque-surrounding glia influences the local transcriptome within plaque-glia niches, we contrasted the glia-high versus glia-low gray matter spots with and without stratification by A β intensity (Fig. 5A). We first estimated TMM-normalized pseudobulk expression of glia-high and glia-low spots for each condition and then applied LMM with FDR correction for all tested genes and contrasts. We detected 230, 241, and 63 DEGs for combined, low A β and high A β conditions (FDR adjusted $p < 0.05$; Fig. 5B-D; Supplementary Table 8), respectively. We detected higher expression of glial markers (*SPARC*, *CD44*, and *HLA-DRA*) in the glia-high spots across all three A β conditions (Fig. 5B, G). Furthermore, we detected lower expression of excitatory and inhibitory neuronal makers (*RORB*, and *PVALB*), synaptic signaling (*SYT2*, *SLC6A17*, *GRM7*, and *GABBR2*), ion channels (*SCN1B*), and neurofilaments (*NEFH/M/L*) in glia-high spots (Fig. 5B, G). We also detected higher expression of translation genes (*EEF1A1*) in glia-high spots (Fig. 5B, Supplementary Fig. 9), consistent with a recent study demonstrating that increased local protein synthesis in microglia serves as a necessary step for phagocytosis [44]. Our GSEA analysis showed a negative enrichment for synapses, exocytic vesicles, neural filaments, and voltage-gated channel activity and positive enrichment for many immune-related terms under all A β conditions (Fig. 5E), in line with the recent high-resolution imaging study demonstrating that reactive glia nets around A β plaques are toxic inflammatory microenvironments in postmortem AD brains and animal models [4].

To assess the independent effects of astrocytes or microglia, we stratified and labeled spots based on the intensities of A-beta, GFAP and IBA1. For example, HHL denotes high A-beta, high GFAP, and low IBA1, respectively. We examined four contrasts: HHL vs. HLL, HLH vs. HLL, LHL vs. LLL, and LLH vs. LLL. After retaining brain sections with at least 50 spots for pseudobulk analysis, the number of available individuals per group was limited: 1 (HHL), 6 (HLH), 5 (HLL), 5 (LHL), 11 (LLH), and 15 (LLL). Due to these limited sample sizes, we detected minimal DEGs, except for LHL vs. LLL contrast, which revealed 133 DEGs. (Supplementary Table 9).

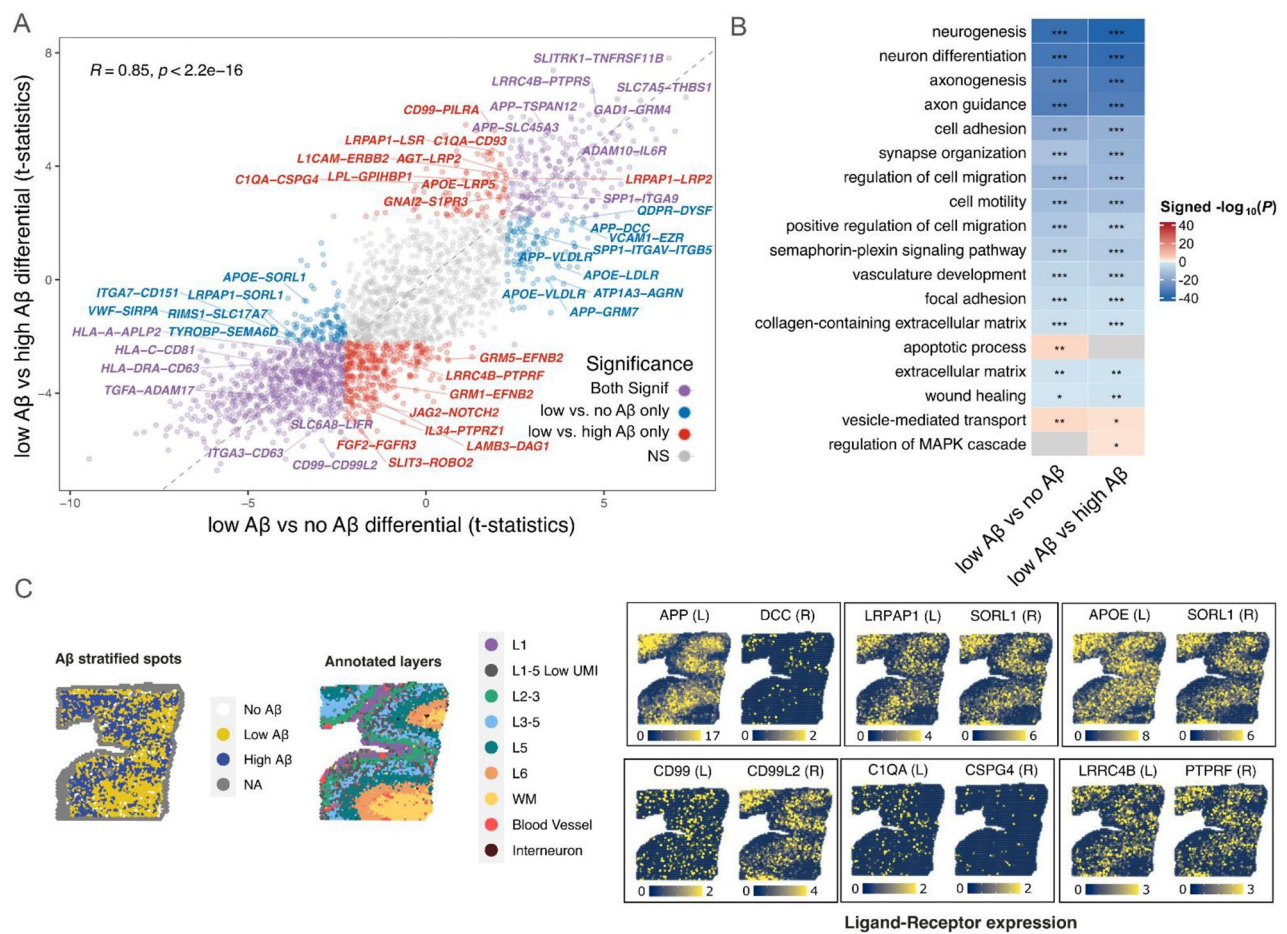


Fig. 4 A β plaques influence intercellular communication. **A** Differential ligand-receptor (LR) interaction genes between low A β vs. high A β (y-axis) and low A β vs. no A β spots (x-axis). Each dot corresponds to an LR pair tested; the axis represents the t-statistics from linear mixed model tests accounting for repeated donors. Significant LR pairs (FDR < 0.05) are highlighted in red (low A β vs. high A β only), blue (low A β vs. no A β only), or purple (significant in both comparisons). **B** Selected Gene Ontology terms enrichment from LR differentially expressed in low A β vs. no A β and low A β vs. high A β . Both ligand and receptor genes were considered for the enrichment analysis. P-values were adjusted using the *g:SCS* method from *gprofiler*. Colors in the heatmap represent the signed $-\log_{10}(\text{adj. } P)$. * $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$. **C** Representative Sect. (17_D) image colored by normalized expression of selected differentially expressed LR gene pairs. All the statistical analyses were performed on gray matter spots only

Depending on the A β conditions, we detected different gene expression changes and geneset enrichment. Under the low A β condition, expression of genes related to lysosome/autophagy (*ATP13A2*) is lower, and expression of genes related to immune (*TRIM22*) and axon guidance/apoptosis (*SEMA3C*) is higher in glia-high spots (Fig. 5E, Supplementary Fig. 9). Under the high A β condition, expression of genes related to phagocytosis (*CD68*), ECM (*SERPINE2*), and complement pathway (*C5*) are higher in the glia-high spots (Fig. 5E, Supplementary Fig. 9A). Our GSEA analysis showed a stronger negative enrichment of synapses, intermediate filament assembly, and voltage channels in the low A β condition, accompanied by stronger positive enrichment of apoptosis, complement

cascades, and cytokine production, supporting a more profound effect of surrounding glia in the low A β condition. Thus, our results indicate that plaque-surrounding glia may elevate neuronal toxicity and synapse degeneration in the aged human brain, especially in low-A β spots. Because inflammatory response [47], complement cascades [48], disturbed extracellular vesicle release [49], and abnormal lysosome-autophagy pathways [50] have been implicated in AD pathogenesis, our results also suggest that the plaque-surrounding glia are implicated in multiple facets of AD neurodegeneration.

In addition to gene ontology (GO), we further applied GSEA with gene signatures of PIG [30], OLIG [30], DAM [31], and DAA [32] and different microglia states [8]

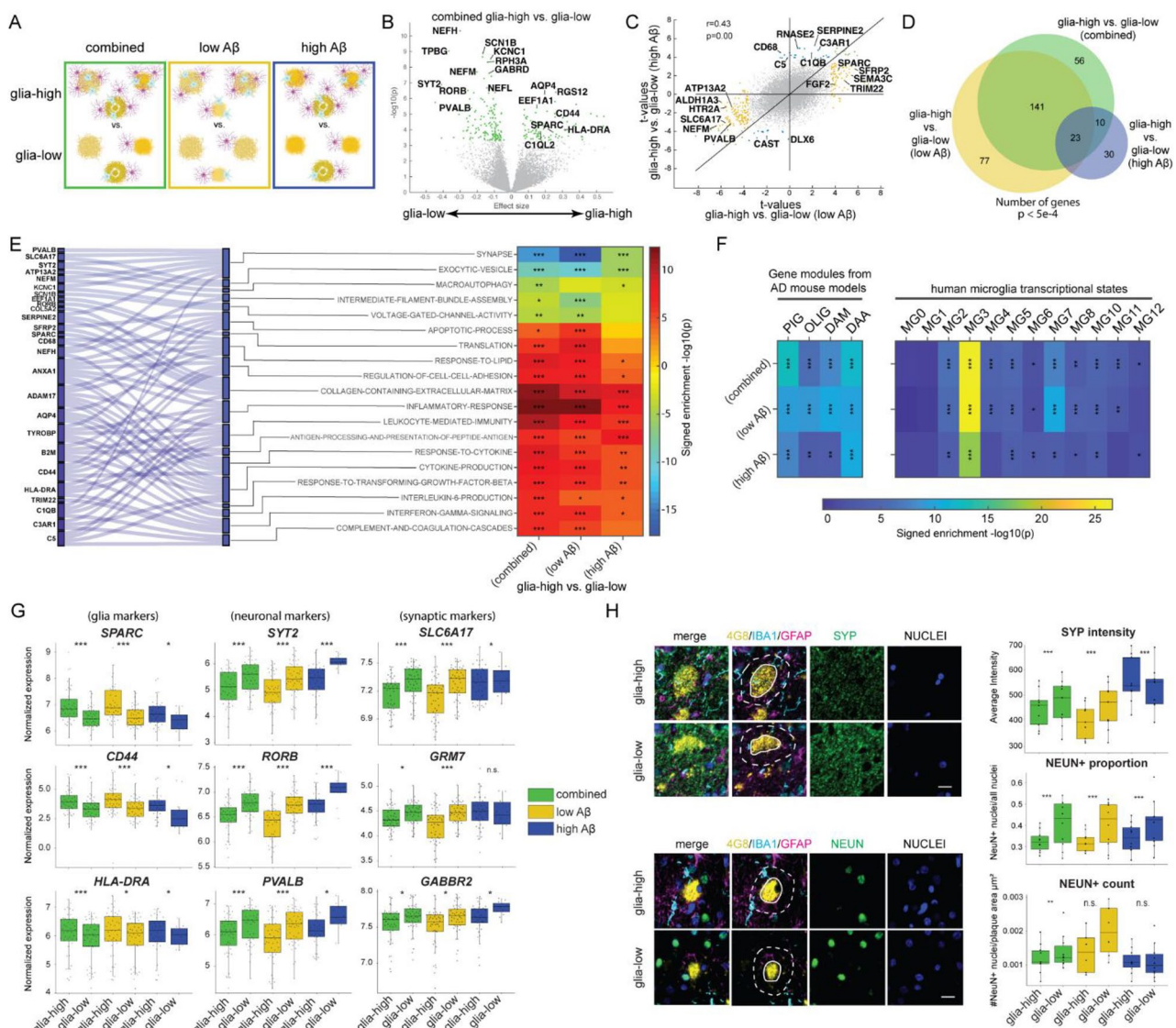


Fig. 5 Differential effects of glia abundance on the local transcriptome. **A** Schematic of the three contrasts tested. A β and glia graphics created with BioRender.com. **B** Volcano plot of glia-high vs. glia-low. Genes with FDR-adjusted $p < 0.05$ in green. **C** Scatterplot of glia effects stratified by low or high A β . Genes with FDR-adjusted $p < 0.05$ in yellow and blue for low and high A β conditions, respectively. **D** Venn diagram on the number of DEGs found across glia contrasts. **E** GSEA enrichment of glia contrasts for GO terms and canonical pathways (* $p < 5e-4$, ** $p < 5e-5$, *** $p < 5e-6$). **F** GSEA enrichment of glia contrasts for A β /glia-relevant genesets (* $p < 5e-2$, ** $p < 5e-3$, *** $p < 5e-4$). PIG³¹: plaque-induced genes, OLIG³¹: oligodendrocyte gene module, DAM³²: disease-associated microglia, DAA³³: disease-associated astrocytes; MG0-12: microglia states in human AD brains⁸. **G** Boxplot of representative genes found for each glia contrast (LMM, FDR-corrected; * $p < 0.05$, *** $p < 5e-4$, n.s. not significant). **H** Images of representative high A β plaques from FFPE tissue sections stained with DAPI (blue), A β (4G8; yellow), IBA1 (cyan), GFAP (magenta), and synaptophysin (SYP; green, top-left panel) or NEUN (green, bottom-left panel). SYP and NEUN display lower-intensity staining surrounding glia-high plaques compared to glia-low plaques (quantification on right). Points represent average values from each of 10 AD individuals quantified in the area surrounding plaques (scale bar = 25 μ m; paired t-test; ** $p < 0.005$, *** $p < 1e-10$, n.s. not significant).

(Fig. 5F). We detected enrichment of PIG, OLIG, DAM, and DAA under both low A β and high A β conditions, indicating an elevated glial response from multiple glial cell types, including microglia, astrocytes, and oligodendrocytes. Indeed, the PIG module has been shown to involve multiple cell types and represent intercellular

crosstalk between astrocytes and microglia [12]. Among the reported human AD microglia states [8], we detected enrichment for almost all MG states except for MG0 (homeostatic) and MG1 (neuronal surveillance) [8]. The MG3 DAM (ribosome biogenesis) state showed the strongest enrichment under all A β conditions. The

enrichment of the MG3 state for glia response is consistent with the positive GO enrichment for translation (Fig. 5E). In addition, the enrichment for MG4 (lipid processing) and MG7 (glycolytic) aligned with the positive GO enrichment for lipid response. Further, the enrichment for MG5 (phagocytic), MG8 (inflammatory II), and MG10 (inflammatory III) are in line with the GO enrichment for immune responses. As such, these results indicate that diverse MG states, especially MG3, are present around the plaques. Taken together, our results suggest that multiple glial cell states arise around plaques and contribute to local immune response.

We then set out to validate the synaptic and neuronal loss in the high-glia spots. To this end, we stained FFPE sections from 10 AD individuals with A β , GFAP, and IBA1 and synaptophysin (SYP) to label synapses or NeuN to label neuronal nuclei (Fig. 5H). Consistent with our ST IHC analysis, 4G8 objects with an ROI scoring in the top 25% for scaled GFAP or IBA1 average intensities were defined as glia-high and the remaining objects as glia-low. The intensity and count measurements were collected from the ROI defined as the 25 μ m ring around the plaque. We then applied the NEBULA model with NeuN count as the response to test the effects of glia-high vs. glia-low on neuron loss. We also examined glia effects on synapse loss by applying linear mixed models with average SYP as the response and within-subject correlations among ROIs from the same subject as a random effect. We detected reduced staining of these neuronal/synaptic markers near glia-high plaques compared with glia-low, especially for low A β plaques (Fig. 5H), in line with our ST data (Fig. 5G). Notably, our result is consistent with a recent ST study showing alterations in GABAergic and glutamatergic signaling corresponding to microglia density in plaque-glia niches in a mouse model of AD [51]. In addition, we stained the brain sections with antibodies against A β , GFAP, IBA1, and CD68 to examine whether glia-high A β spots are associated with increased neuroinflammation. Our result shows a higher abundance of CD68 protein near glia-high than glia-low plaques (Supplementary Fig. 9C).

Inferred intercellular communication suggests synaptic degeneration in glia-high plaque spots

We again applied NICHES to estimate differences in microenvironments centering around each ST spot. We first contrasted glia-high versus glia-low to identify differentially expressed LR interactions under low A β , high A β , and combined conditions (Supplementary Table 8). We observed a positive correlation ($R = 0.41$) between the low A β and high A β conditions. The expression of LR interactions related to synaptic pathways (*RPH3A-NRXN1*, *DSCAM-DCC*, *GNB3-GABBR2*, *RIMS1-SLC17A7*, and *CXCR16-GRM7*) was

downregulated in the glia-high spots across all A β conditions (Fig. 6A-C). In the high A β condition, we also detected an upregulation of genes related to immune (*CLU-TREM2*, *PLXNA1-TREM2*, *APOE4-LDLR* and *GFAP-APLN*) and extracellular cell matrix (*SPARC-ENG*), vesicle transport (*APP-TSPAN15*) as well as a downregulation of TGF- β 1 signaling (*TGFB1-ITGAV*). Concordantly, our GSEA analysis on differentially expressed LR interactions revealed more substantial LR differences in the high A β condition, including increased autophagy as well as reduced cell adhesion and myelination (Fig. 6D). Because NICHES measures LR interactions of the microenvironment surrounding each ST spot, the more pronounced functional implications in the high A β condition may reflect a culmination of progressive plaque development, glial activation, and neurodegeneration.

ST glial response to A β plaques recapitulated by co-cultured microglia-like cells upon A β oligomer treatment

Studies have shown that A β plaques are potential reservoirs of oligomeric A β , often forming a "halo" around A β plaques [52, 53]. To examine whether the glial transcriptional signature detected by ST reflects response to A β oligomers and to what extent microglia or/and astrocyte mediate this response, we established a co-culture system encompassing iPSC-derived microglia-like cells (iMGLs), astrocytes, and neurons to mimic the multicellular nature of the brain tissue (Fig. 7A). We performed immunofluorescent staining 72 h after co-culture and detected IBA1 + iMGLs, GFAP + astrocytes, and DCX + neurons (Fig. 7B). We then treated co-cultures with A β oligomers for 24 h and harvested the cells for scRNA-seq using the 10X Genomics Chromium platform.

We acquired scRNA-seq data from 13,740 cells under the A β untreated and treated conditions. After standard data processing and filtering, we applied unsupervised clustering on the first 15 principal components to group the cells (Supplementary Fig. 10 C). We identified 5 major cell clusters that we annotated based on classical markers as iMGLs, astrocytes, neurons, cycling neuroglial progenitors, and other progenitors (Fig. 7C-D). To examine the A β oligomers' effect, we analyzed differential gene expression for neurons, astrocytes, and iMGLs, and identified 199, 58, and 271 A β DEGs, respectively (Supplementary Table 10). We then performed GSEA on the glia-high versus glia-low contrasts from ST data with the DEGs upregulated upon A β -oligomer treatment as the geneset. We detected a positive enrichment of the ST glial response in iMGLs. However, no enrichment was observed in astrocytes and neurons (Fig. 7E). The DEGs in iMGLs included many immune, complement cascade, and phagocytosis genes, such as *HLA-DRA*,

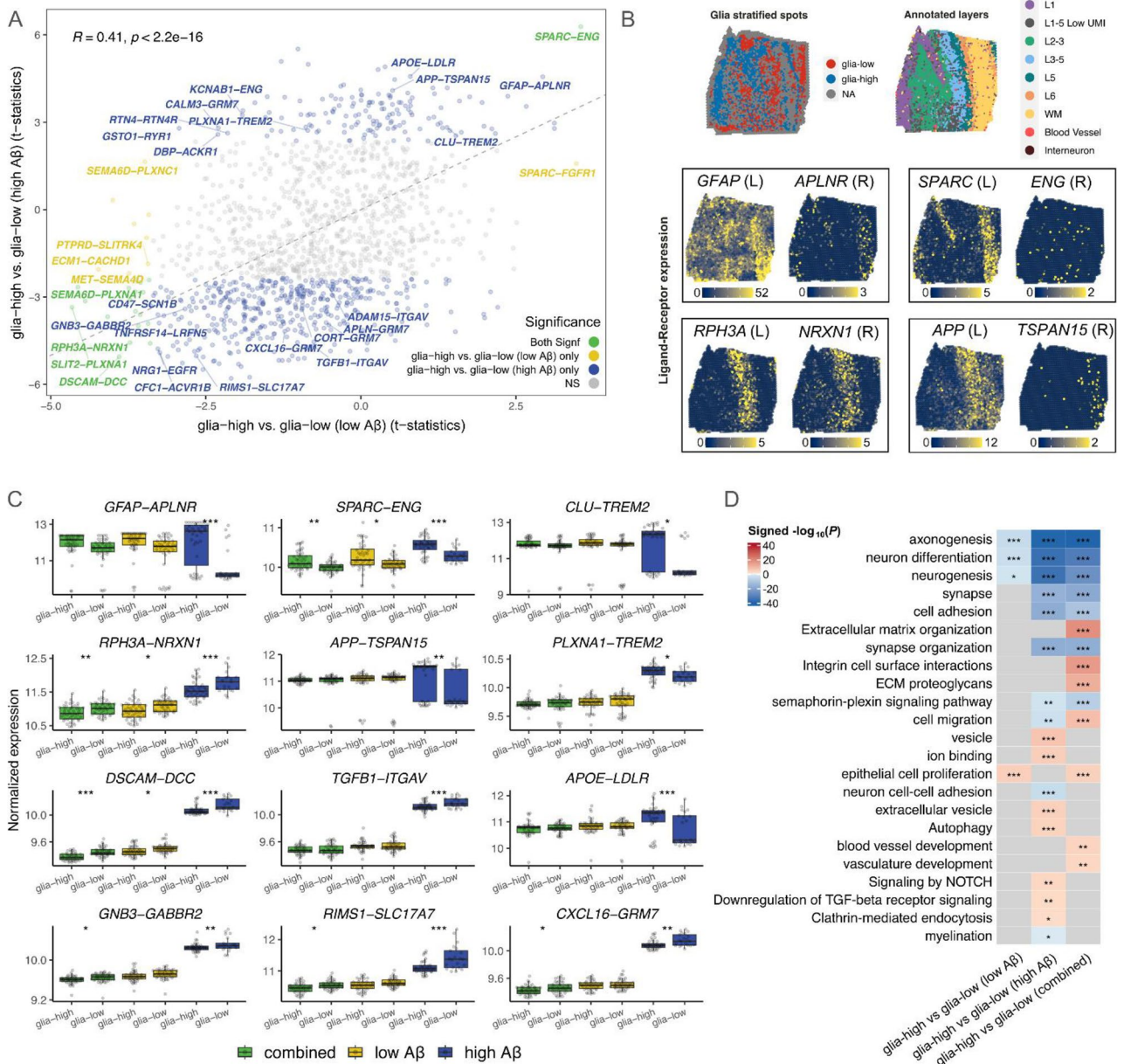


Fig. 6 Plaque-surrounding reactive glia modifies intercellular communication. **A** Differential ligand-receptor (LR) genes between glia-high vs. glia-low under High A β spots (y-axis) and glia-high vs. glia-low under low A β spots (x-axis). Each dot corresponds to an LR pair tested; the axis represents the t-statistics from linear model tests accounting for repeated donors. Significant LR pairs (FDR < 0.05) are highlighted in blue (glia-high vs. glia-low under low A β only), yellow (glia-high vs. glia-low under high A β only), or green (significant in both comparisons). **B** Example Sect. (15_D) colored by normalized expression of selected differentially expressed LR gene pairs. All the statistical analyses were performed on gray matter spots only. **C** Normalized expression levels by stratified spot groups for selected differentially expressed LR pairs. (*p ≤ 0.05, **p ≤ 0.01, ***p ≤ 0.001). **D** Selected Gene Ontology terms enrichment from LR differentially expressed between glia-high vs. glia-low under low A β , high A β , and combined were considered for the enrichment of both ligand and receptor genes. P-values were adjusted using the gSCS method from gprofiler. Colors in the heatmap represent the signed -log₁₀(adj. P) (*p ≤ 0.05, **p ≤ 0.01, ***p ≤ 0.001)

CD68, C4ORF48, CXCL16, ITGAX, and CIQB (Fig. 7F). In agreement, GSEA on the co-culture data revealed enrichment for increased immune response, reduced cell metabolism, and cytokine release, aligning with the

functional pathways of the ST glial response (Supplementary Table 10). Thus, our results suggest that the molecular effect of plaque-surrounding glia in tissue may result from the A β oligomers residing in plaques. Although

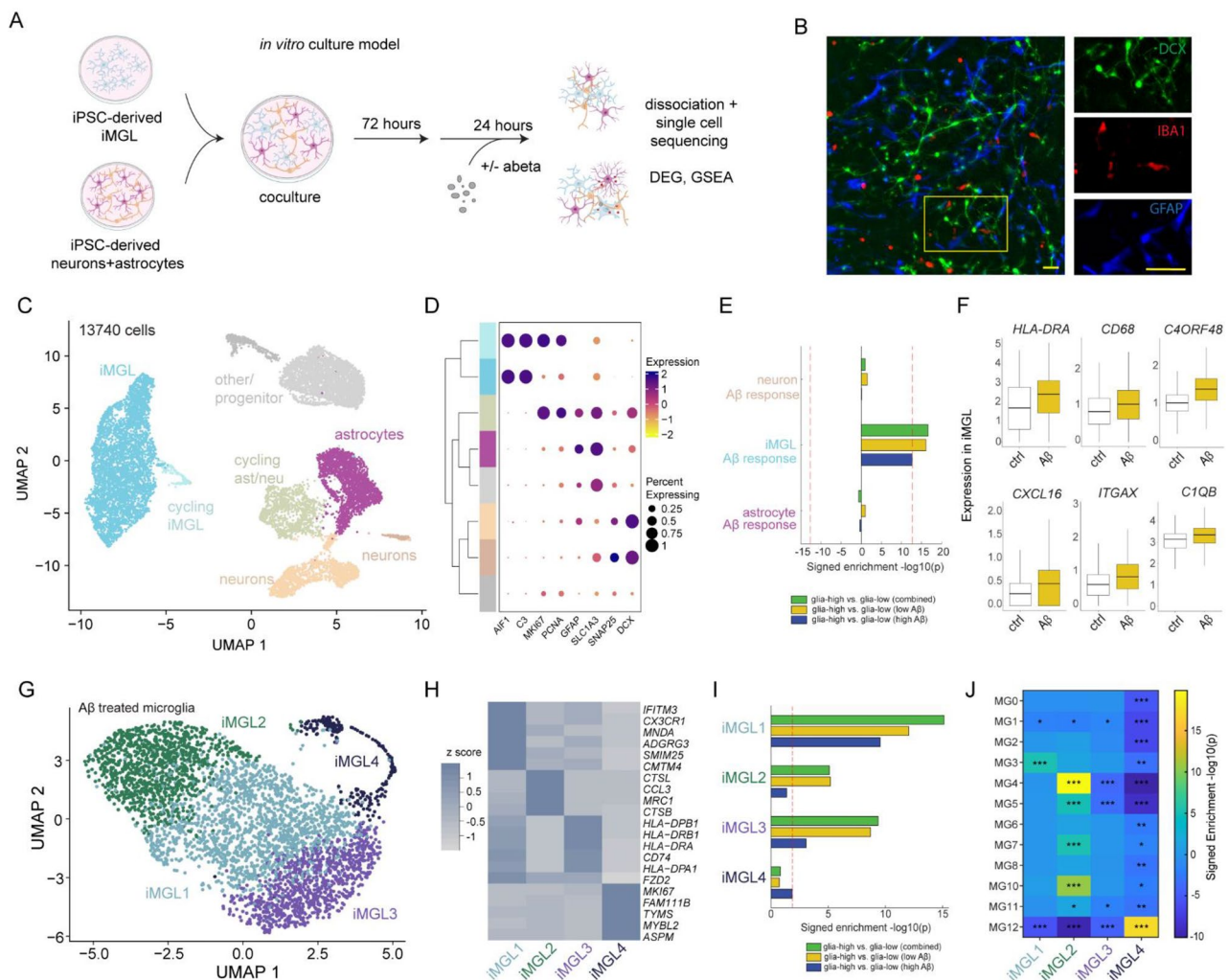


Fig. 7 A β oligomer treatment of co-cultured iMGL partially recapitulates ST glia response to A β plaques. **A** Schematic of in vitro co-culture experiments. Graphics created with BioRender.com **(B)** Representative immunofluorescence for DCX (neuronal), IBA1 (iMGL) and GFAP (astrocyte) proteins in co-culture models, scale bar: 100 μ m, inset: 50 μ m. **C** UMAP visualization of cells from control and A β oligomer-treated cultures, colored by annotated cell type clusters. **D** Dotplot of cluster/cell-type enriched genes. **E** GSEA enrichment of A β -induced genes for ST glial response by cell type. Red dotted line corresponds to FDR-threshold. **F** Boxplots of genes differentially expressed in iMGL upon A β treatment. **G** UMAP visualization of A β oligomer-treated iMGL, colored by subtype cluster. **H** Heatmap of the top 5 differentially expressed gene markers per iMGL subtype cluster. **I** GSEA enrichment of A β -treated iMGL cluster DEGs (Supplementary Table 11) for ST glial response. Red dotted line corresponds to FDR-threshold. **J** GSEA enrichment of A β -treated iMGL cluster (vs. control) DEGs (Supplementary Table 12) for reported MG state genesets in human AD brains[8] (* $p < 5e-2$, ** $p < 5e-3$, *** $p < 5e-4$)

both astrocytes and microglia are present around plaques, our results support the idea that microglia may play a primary role in mediating glial response. Consistently, the recent spatial profiling of plaques in mouse AD brains and high-resolution imaging on the human brain reveal that microglia closely surround plaques and astrocytes are more distant from the plaques [4, 13].

We then further divided the A β -treated iMGLs into 4 subclusters (iMGL1-4; Fig. 7G, 7H) and applied GSEA on the glia-high vs glia-low contrasts from ST data with DEG markers of each iMGL subcluster as genesets

(Supplementary Table 11). iMGL1, iMGL2, and iMGL3 showed different degrees of enrichment of the ST glial response (Fig. 7I). Specifically, iMGL1 demonstrated enrichment under both A β -high and A β -low conditions, whereas iMGL2 and iMGL3 showed enrichment primarily in the A β -low condition (Fig. 7I). These results suggest that the responses of iMGL1-3 to oligomeric A β , especially iMGL1, constitute a major element of the ST glial response. To understand which MG states the iMGL subclusters resemble in human AD brains, we applied GSEA on the transcriptomic signature of each iMGL subcluster

with markers of 12 reported MG states as genesets. (Supplementary Table 12) [8]. Notably, iMGL1 is positively enriched for the MG3 DAM state, which is the most enriched state in the ST glial response (Fig. 5F). iMGL2 is enriched for MG4 (lipid processing), MG5 (phagocytic), MG7 (glycolytic), and MG10 (inflammatory III) (Fig. 7J), which is also in line with the enriched states in the ST glial response (Fig. 5F). Thus, our results show that the MG states in the AD brain can be recapitulated by co-cultured microglia upon A β treatment. This suggests multiple reported MG states, especially MG3, shape the local glial response to A β deposition in human AD brains.

Discussion

While A β plaques and reactive glia have long been known to be associated with neurodegeneration, their interaction effects on local transcriptomes have not been extensively studied using ST. Here, we report paired ST-IHC measurements across 78 samples from 21 ROSMAP individuals. This large sample size enables statistical inference of A β and glial effects that are not merely observed in a limited sample size. By coupling ST readouts with IHC features of A β plaques and glial cells, we identify genes, pathways, and cell communication differences associated with distinct plaque-glia niches, shedding light on the cellular and molecular environments of plaque-related neurodegeneration. We demonstrate that low-A β areas, enriched for diffuse plaques, are more neurotoxic than high-A β areas and an increased abundance of reactive glia in plaque-glia niches exacerbates neurodegeneration. The glial response genes we identified by ST are enriched for multiple MG states identified by snRNA-seq of human AD brains. Importantly, exposure of co-cultured iMLCs to A β oligomers recapitulates the ST glial transcriptional response, as well as the reported MG states in human AD brains.

Low versus high A β spots

By contrasting low A β and high A β spots, we detected transcriptomic changes indicative of increased synaptic degeneration in the low A β spots. First, our DEG and GSEA analysis identified genes and functional pathways related to synaptic degeneration. Second, our NICHES analysis reveals more drastic LR differences related to neuronal and synaptic degeneration in the low A β spots. Third, our IHC validation experiment shows increased cleaved caspase-3 puncta co-localized with NeuN + nuclei and total nuclei near low A β plaques compared to high A β plaques. Thus, all our results point toward more detrimental effects in low A β spots. It is possible that we detected increased apoptosis and neuronal degeneration signatures in low A β spots

because vulnerable neurons near high A β spots have already undergone cell death. This is consistent with our finding that the number of NeuN + nuclei is lower surrounding high A β plaques after normalizing by plaque area (Fig. 5H; compare blue and yellow bars of boxplot). Our result resonates with a recent study showing that A β can accumulate into plaque depositions inside damaged neurons, subsequently contributing to extracellular A β plaque formation [54], suggesting that the neuronal damage observed in AD could occur at the early stage of plaque development when A β loads are relatively low.

This observation is also concordant with a recent label-free vibrational imaging study showing that early-stage plaques consist mainly of toxic A β oligomers and protofibrils, whereas late-stage plaques primarily contain A β fibrils [45]. Moreover, a recent electron cryomicroscopy (cryo-EM) study demonstrated two types of A β 42 filaments in human brains. Type I filaments are mostly in sporadic AD cases with abundant compact plaques, and Type II filaments are in familial AD cases with abundant diffuse plaques [55]. Considering the general severity of A β deposition in familial AD, this result also supports more neuronal toxicity of diffuse plaques.

Another interesting observation gleaned from contrasting low A β and high A β spots is that surrounding glial cells influence the local transcriptome. For example, we detected an upregulation of ECM pathways in the glia-low condition and an upregulation of immune and apoptotic pathways in the glia-high condition (Fig. 3E, 3G). Since low-A β plaques represent the early phase of plaque development, this divergence could represent differential coping mechanisms in the glia-high versus glia-low spots. In the glia-low condition, the upregulation of *ICAM* and *DAB1* (Supplementary Fig. 8A) might be beneficial because *ICAM1* can protect neurons against A β by blocking NF- κ B signaling [56], and *DAB1* activation by Reelin (*RELN*) is known to modulate synaptic activity [57, 58]. In the glia-high condition, the upregulation of *ADAM10* and *SFRP2* could also be compensatory. *ADAM10* in astrocytes has been shown to cleave the APP protein and prevent A β generation [59], and secreted frizzled-related proteins (SFRPs) can bind to A β and prevent protofibril formation, thereby modulating astrocyte-microglia crosstalk in neuroinflammation [60–62]. These results indicate that plaque-glia niches might utilize different mechanisms to cope with A β -related cell damage depending on the context of glial activation.

The effects of plaque-surrounding glia on the local niche

To examine the effect of surrounding glia in the local plaque-glia niches, we contrasted glia-high versus glia-low spots in the low A β and high A β conditions. Our analysis indicates that the surrounding glia may mount a

robust inflammatory response and exacerbate neurodegeneration, especially in the low A β spots. Our IHC validation experiments further validated this result, showing reduced neuronal/synaptic and increased inflammation marker expressions near glia-high plaques compared with glia-low plaques (Fig. 5H). The neuronal degeneration in glia-high spots is consistent with recent studies. First, microglia contain and accumulate A β aggregates, which could lead to microglial death and the release of intracellular A β aggregates, contributing to plaque expansion and neurodegeneration [63]. Indeed, eliminating microglia in AD mouse models stops plaque growth and reverses AD-related transcriptomic changes [64]. Second, microglia activate the NLRP3 inflammasome, which subsequently drives the formation and release of ASC specks (apoptosis-associated speck-like protein containing a CARD), increasing A β oligomer formation, immune response, and neuronal damage [65]. Lastly, dysregulated crosstalk between microglia and astrocytes also escalates neuroinflammation and cell death. For example, A1 astrocytes can be induced by activated microglia, which can induce neuronal death and synaptic loss; in accordance, the presence of A1 astrocytes is reported in the human AD brain [66]. Thus, our results support a harmful and non-resolving inflammatory response in glia-surrounding plaques, resulting in elevated neuronal degeneration.

Furthermore, our GSEA analysis shows that the ST glial response is enriched for signatures of reported mouse gene modules of plaque-induced genes (PIG), oligodendrocyte (OLIG) response, disease-associated microglia (DAM), and disease-associated astrocytes (DAA), as well as different microglia (MG) states identified in human AD brains, indicating that multiple glial cell states arise around plaques and contribute to local immune response. Among all the enriched MG states, MG3 is the most enriched in the ST glial response. This MG state has been shown to resemble mouse DAM and expresses inflammatory genes enriched for cytokine production, antigen presentation, and microglial activation [8]. The strong enrichment of the MG3 state provides a potential therapeutic cell state target for fine-tuning the immune response in human AD brains. In addition, our GSEA analysis also shows a greater leukocyte-mediated immunity in glia-high spots across all A β conditions (Fig. 5E), which is unexpected as the innate immune response (MGs and astrocytes) has been considered the primary culprit of neuroinflammation in AD. Nevertheless, our result aligns with recent studies demonstrating that microglia can mediate T cell infiltration in the mouse brain with tauopathy, and the number of T cells was correlated with the extent of neuronal loss. An increased number of T cells was also found in the AD postmortem brains [67]. In addition, CD163-positive

monocyte-derived cells were shown to infiltrate brain parenchyma near A β plaques, contributing to plaque-associated myeloid cell heterogeneity [68]. Therefore, an immune hub involving innate and adaptive responses may be present in aged human AD brains, especially in the plaque-glia niches.

***In vitro* validation of ST glia response**

The iPSC-derived microglia have been shown to recapitulate disease microglia states upon exposure to AD-related stimuli and master regulators [8, 69]. To disentangle the glia effect observed in brains, we established an iPSC-derived multicellular culture system that emulates the cellular interactions in the brain tissue microenvironment. We show that iMGLs, upon A β -oligomer treatment, but not astrocytes, mirror the ST glial response. Among iMGL subclusters, iMGL1-3 clusters demonstrate an enrichment for glial response. Notably, iMGL1 most resembles MG3 DAM state, and iMGL2 resembles multiple MG states in human AD brains. These results further extend our ST analysis of the glial response, indicating that multiple MG states mediate glial response and microglia's response to toxic A β oligomers initiates glial activation in the plaque-glia niches. Although we did not detect enrichment for other cell types, the crosstalk among astrocytes, neurons, and microglia may contribute to microglia's strong molecular response. Indeed, crosstalk between astrocytes and microglia has been shown to increase the degradation of A β aggregates in iPSC-derived cultures [70]. Furthermore, given that we only treated co-culture for 24 h, it is also possible that prolonged treatment may trigger a pronounced response in neurons and astrocytes directly or indirectly. Nonetheless, the positive enrichment of the ST glial response in iMGLs supports a key role of microglia in driving glial response in plaque-glia niches.

Conclusions and future directions

Our study demonstrates how advanced ST technology can be effectively combined with classical histology to address questions that neither approach can answer alone. This integrated workflow and computational framework offer a novel and accessible method for investigating pathological heterogeneity in AD and other neurodegenerative disorders. By coupling ST data with traditional A β IHC staining, we provide new insights into the brain's cellular and spatial context, linking the abundance of plaques and nearby reactive glia to neuronal vulnerability, synaptic degeneration, and inflammatory responses. Our study also highlights the power of integrating ST, snRNA-seq, and human cell models to decipher the sequential events and cell states in plaque-glia niches, indicating that diverse microglial activation states arise around A β plaques and that microglial response

to toxic A β oligomers initiates glial activation. Overall, our research lays the groundwork for future pathology genomics, opening new avenues for addressing pathological heterogeneity in neurodegenerative diseases. Future studies with increased sample size and diversity, high-resolution spatial profiling platforms, the ability to work with fixed/archived tissues, and co-detection of proteins on the same section will address many technical challenges we encountered and enable finer delineation of cell-type-specific responses to distinct pathologies.

Limitations

There were several limitations of this study. First, our ST samples were collected from aged female individuals (mean age of 92). Therefore, whether our results can generalize to younger age groups or males remains to be determined. Second, because A β spots primarily came from AD cases, we were underpowered to assess the intersection of AD status with the detected transcriptional signatures in plaque-glia niches. Third, as the Visium platform is at ~55- μ m resolution and both astrocytes and microglia accompanied most plaque-containing spots, we were also underpowered to analyze astrocyte- and microglia-specific transcriptional responses in plaque niches. Fourth, due to frequent freezing artifacts of postmortem frozen tissue and the large size of this dataset (258,986 spots), we could not definitively categorize some of the plaques as the diffuse, compact, or dense core; thus, we used no, low and high A β to group the ST spots instead. Fifth, because of the relatively low resolution of the Visium platform and the limitation to choosing three proteins for IHC, we focused on extracellular A β pathology and surrounding glia to investigate the A β -glia niches. Future studies will investigate molecular responses to intracellular pathologies, such as neurofibrillary tangles, using high-resolution ST platforms. Finally, although our A β -treated microglia-like cells recapitulate major immune pathways and map partially to ex vivo microglial states, cultured microglia are known to diverge transcriptionally from in vivo microglia due to the absence of brain-specific environmental cues [71, 72].

Abbreviations

AD	Alzheimer's disease
NCI	No cognitive impairment
A β	Amyloid-beta
IHC	Immunohistochemistry
ST	Spatial Transcriptomics
ROSMAP	Religious Orders Study and Memory and Aging Project
FFPE	Formalin-fixed paraffin-embedded
DEGs	Differentially expressed genes
FDR	False discovery rate
RIN	RNA Integrity Number
GSEA	Gene set enrichment analysis

PIG	Plaque-induced genes
DAM	Disease-associated microglia
DAA	Disease-associated astrocytes
MG	Microglia
IMGLs	IPSC-derived microglia-like cells

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s44477-025-00002-z>.

Below is the link to the electronic supplementary material. Supplementary file1 (XLSX 149 KB)

Supplementary file2 (XLSX 754 KB)

Supplementary file3 (XLSX 610 KB)

Supplementary file4 (XLSX 17 KB)

Supplementary file5 (XLSX 18 KB)

Supplementary file6 (XLSX 99 KB)

Supplementary file7 (XLSX 40 KB)

Supplementary file8 (XLSX 1665 KB)

Supplementary file9 (CSV 57 KB)

Supplementary file10 (XLSX 3431 KB)

Supplementary file11 (XLSX 2087 KB)

Supplementary file12 (XLSX 3568 KB)

Supplementary file13 (XLSX 6085 KB)

Supplementary file14 (DOCX 6.62 MB)

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Author contributions

D.R.A., S.D.T., and D.J.F. performed sample prep and ST experiments. J.X. performed ST data and scRNA-seq data processing. B.N., R.V., D.R.A., N.K., A.I., K.L., C.G. and S.T. performed data analysis. N.K. and D.S. performed cell culture experiments. H.V. and D.S. performed immunohistochemistry experiments. D.R.A., B.N., R.V., and Y.W. wrote the manuscript with input from all co-authors. D.A.B. runs the parent ROSMAP study. Y.W. conceived and supervised this study. All authors reviewed and approved the final manuscript.

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Data availability

Sample/spot information, differentially expressed genes, differentially expressed ligand-receptors, gene modules, ST, and scRNA-seq cluster marker genes are provided in the Supplementary Information. Raw and normalized count matrices of the Spatial Transcriptomics and ST-paired IHC image data are available at Synapse under accession code syn53141470.

Declarations

Ethics approval and consent to participate

Participants were from two ongoing cohort studies conducted at Rush Alzheimer's Disease Center: The Religious Orders Study (ROS) and the Rush Memory and Aging Project (MAP). Each study was separately approved by a Rush University Medical Center Institutional Review Board.

Consent for publication

Written informed consents were collected from all the participants and an Anatomic Gift Act was signed by all ROS and MAP cohorts for brain donation.

Competing interests

The authors declare no competing interests.

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